Wideband RF/Photonic Beamforming Simulator

System-Level Study in MATLAB and Ansys Lumerical INTERCONNECT

George Boutsikas

October 2026

TABLE OF CONTENTS

PART 1: INTRODUCTION 3
System Model and Assumptions 3
PART 2: MATLAB RF BEAMFORMING ANALYSIS 4
Phase-Shifter Beamforming (PS.m) 4
Beam Squint (beam_squint.m) 5
Spatial Sampling and Grating Lobes (spatial_sampling.m) 7
Amplitude Weighting (weighting.m) 8
Multibeam Beamforming (multibeam.m) 9
True-Time-Delay Beamforming (TTD.m) 11
Wideband PS vs TTD Sweeps (worst_cases.m) 12
RF Non-Idealities (RF_nonidealities.m) 13
PART 3: ANSYS INTERCONNECT PHOTONIC TTD 16
Ideal Optical Model (set_ideal.lsf) 16
Non-Ideal Optical Model (set_non-ideal.lsf) 16
Thermal + Noise Case (set_thermal.lsf) 17
Photonic TTD Network (beamforming_project.icp) 17
Response Extraction and Validation (extract_export.lsf) 18
PART 4: END-TO-END MATLAB - INTERCONNECT COMPARISON 22
End-to-End Comparison (end_to_end_comparison.m) 22
End-to-End Quantitative Comparison 28
PART 5: LIMITATIONS & PROJECT SUMMARY 29
Limitations 29
Project Summary 29
PART 6: REFERENCES 30

ABSTRACT

This project presents a system-level comparison of phase-shifter and True-Time-Delay beamforming for an 8-element uniform linear array centered at 10 GHz, followed by an optical TTD implementation in Ansys Lumerical INTERCONNECT. MATLAB is used to analyze steering, element spacing, sidelobes, amplitude weighting, beam squint, multibeam operation, bandwidth limits and RF non-idealities. INTERCONNECT is then used to model the 8-channel optical network, which is characterized through its complex transfer functions $H_n(f)$, and the extracted channel responses are imported back into MATLAB for an end-to-end comparison. For a 30° target steering angle across approximately 7.5–12.5 GHz, the fixed phase-shifter reaches a maximum pointing error of 11.47°, compared with 0.59° for the single randomized non-ideal photonic TTD realization and 1.31° for the separate thermal + noise stress case. The results show the main advantage of TTD for wideband steering, while also showing the insertion loss, channel imbalance and delay error introduced by practical system-level photonic implementations.

PART 1: INTRODUCTION

In this project, the fundamentals of RF and optical beamforming have been presented. The main objective is to compare fixed phase-shifter and True-Time-Delay architectures in a wideband phased array and then connect the RF model to a photonic TTD implementation. The results obtained from both the MATLAB and Ansys INTERCONNECT simulations are shown and explained, while the individual project files are kept in the section titles so that the report remains directly connected to the repository. [1]-[5]

The first part of the project contains an analysis of ideal phase-shifter and True-Time-Delay architectures for an 8-element phased array at a target frequency of 10 GHz in MATLAB. The second part contains a targeted implementation of the optical modulation and delay network for the 8 elements, while the third combines the two into an end-to-end system-level RF/photonic beamforming model. A phase shifter fixes the element phase progression at the design frequency, whereas a true time delay produces a phase that changes linearly with frequency, which is the main reason the two architectures behave differently over wide bandwidth.

System Model and Assumptions

The MATLAB model assumes an ideal uniform linear array with the steering angle measured from broadside. Mutual coupling and full-wave feed effects are not included in the wideband calculations. After the first total-factor example, the remaining MATLAB plots use only the array factor so that the beamforming effects can be isolated.

Table 1: Main simulation parameters

Parameter	**Value / Assumption**
Array geometry	8-element uniform linear array (ULA)
Center frequency	f_0 = 10 GHz
Element spacing	d = $\lambda_0/2$ = 15 mm
Visible angular region	-90 degrees to +90 degrees
Default excitation	Uniform amplitude; progressive phase or delay used for steering
Wideband RF sweep	5-50% fractional bandwidth; steering angles 0-60 degrees
End-to-end frequency range	Approximately 7.5-12.5 GHz
Optical carrier	193.1 THz, 0 dBm
Waveguide indices	n_g = 4.2, n_{eff} = 2.39
Monte Carlo model	5000 trials; independent Gaussian per-element errors, σ_φ = 20° and σ_A = 0.2

PART 2: MATLAB RF BEAMFORMING ANALYSIS

Phase-Shifter Beamforming (PS.m)

The first file demonstrates the basics of beamforming for a uniformly spaced and uniformly weighted array. Steering is introduced through a progressive phase shift between adjacent elements. The main relations used are

$$\Delta\varphi = k \cdot d \cdot \sin(\theta_0)$$

$$\psi = \pi \cdot (d/\lambda) \cdot \sin(\theta) - \Delta\varphi/2$$

$$AF(\theta) = \sin(N\psi) / [N \sin(\psi)]$$

The elements are equally weighted in amplitude, while the progressive phase term is what produces steering. The total normalized patterns, combining the Array and Element Factor, are presented in both Cartesian and polar coordinates for target angles of 0, 20 and 40 degrees. In addition, for each target angle, the Half-Power Beamwidth is calculated both with the approximation

$$HPBW \approx 0.886 \cdot \lambda / [N \cdot d \cdot \cos(\theta_0)]$$

and through a numerical -3 dB crossing calculation. This provides a first validation of the ULA implementation against the expected array behavior. [1], [2]

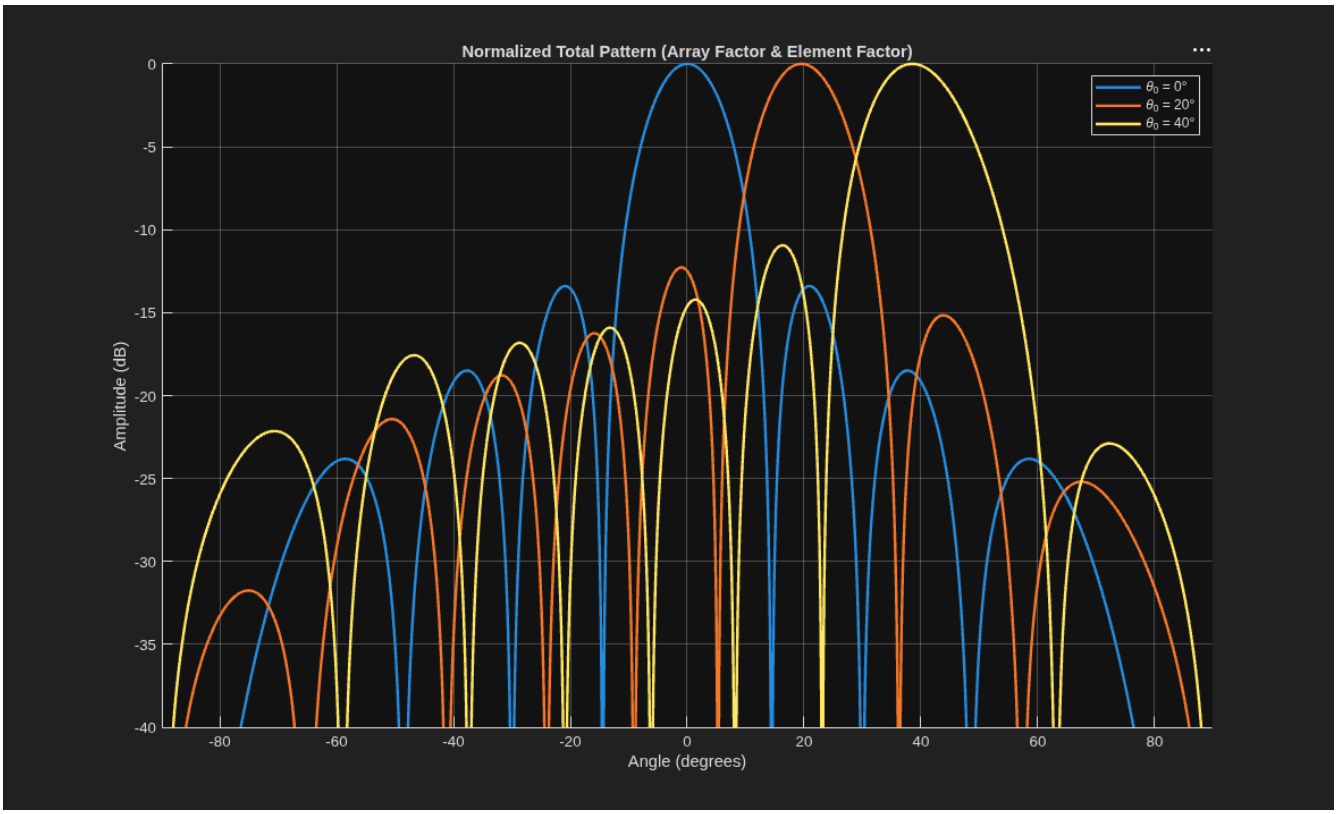


Figure 1: Normalized total factor for different steering angles

From this point on, the MATLAB files contain only the array factor in the calculations, for easier comprehension of the graphs and to isolate the beamforming behavior from the assumed element pattern.

Beam Squint (beam_squint.m)

The second file contains an in-depth analysis of the main drawback of the phase-shifter architecture, beam squint. Firstly, the beam shift is calculated for ratios of operating frequency to target frequency between 0.7 and 1.3. For a phase progression designed at f_0, the analytical beam direction is [3]

$$\theta_{PS}(f) = \sin^{-1}[(f_0/f) \cdot \sin(\theta_0)]$$

$$\Delta\theta(f) = \theta_{PS}(f) - \theta_0$$

$$\text{valid only when } |(f_0/f) \cdot \sin(\theta_0)| \leq 1$$

Because of the asymmetric relation for the beam angle, frequencies lower than the target frequency produce a larger angular shift compared with the corresponding frequency increase above f_0. The effect also becomes stronger as the steering angle moves farther away from broadside.

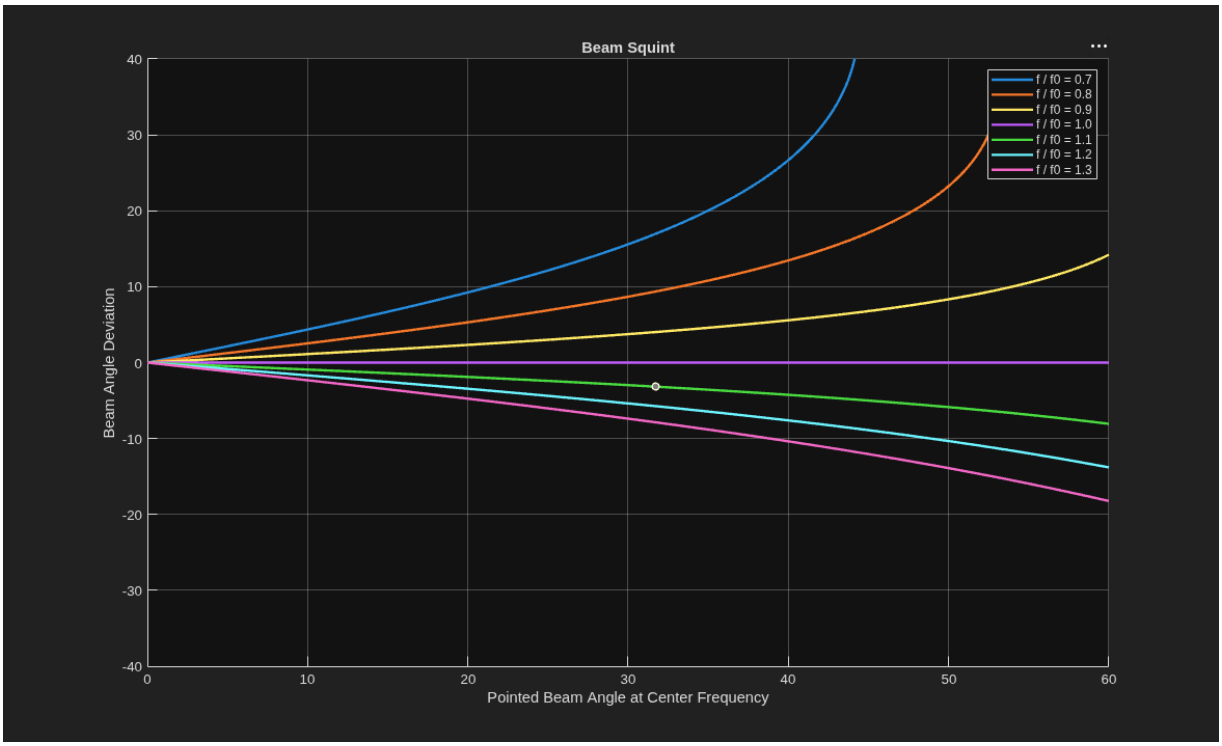


Figure 2: Beam Squint for different frequencies

The second plot shows the comparison between beam direction and frequency. The close agreement between the analytical and numerically extracted peak angles is used as a validation that the MATLAB model is implementing the expected beam squint relation.

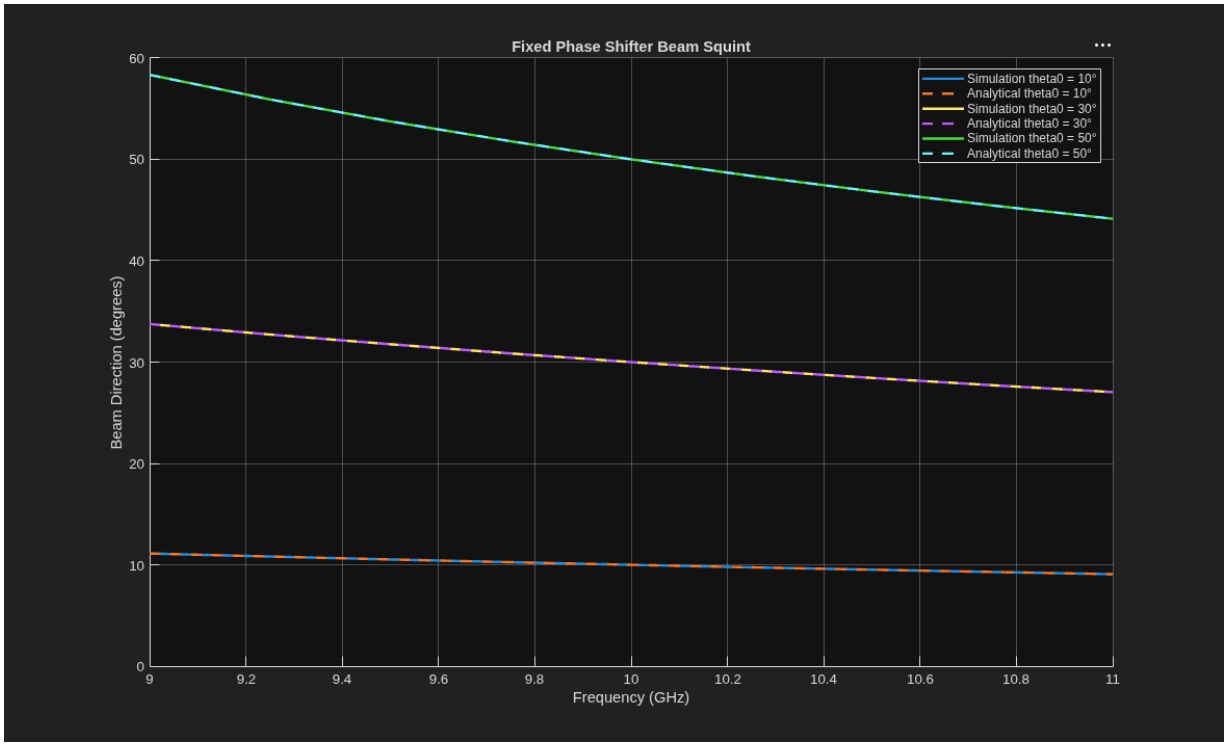


Figure 3: Simulated vs Theoretical beam direction for different frequencies

Spatial Sampling and Grating Lobes (spatial_sampling.m)

The third file presents the differences in the beamforming result for different ratios of element spacing to wavelength. While the higher ratios produce narrower peaks, for ratios of 0.75 and 1 a grating lobe appears at approximately -43.7 and -20 degrees respectively. The grating-lobe positions are calculated from

$$\theta_m = \sin^{-1}[\sin(\theta_0) + m \cdot \lambda / d],$$

for integer m, whenever the arcsin argument lies between -1 and 1. The calculated plots are both Cartesian and polar. This is the spatial-sampling interpretation of grating lobes: increasing d/lambda can create additional visible solutions with full array gain. [3]

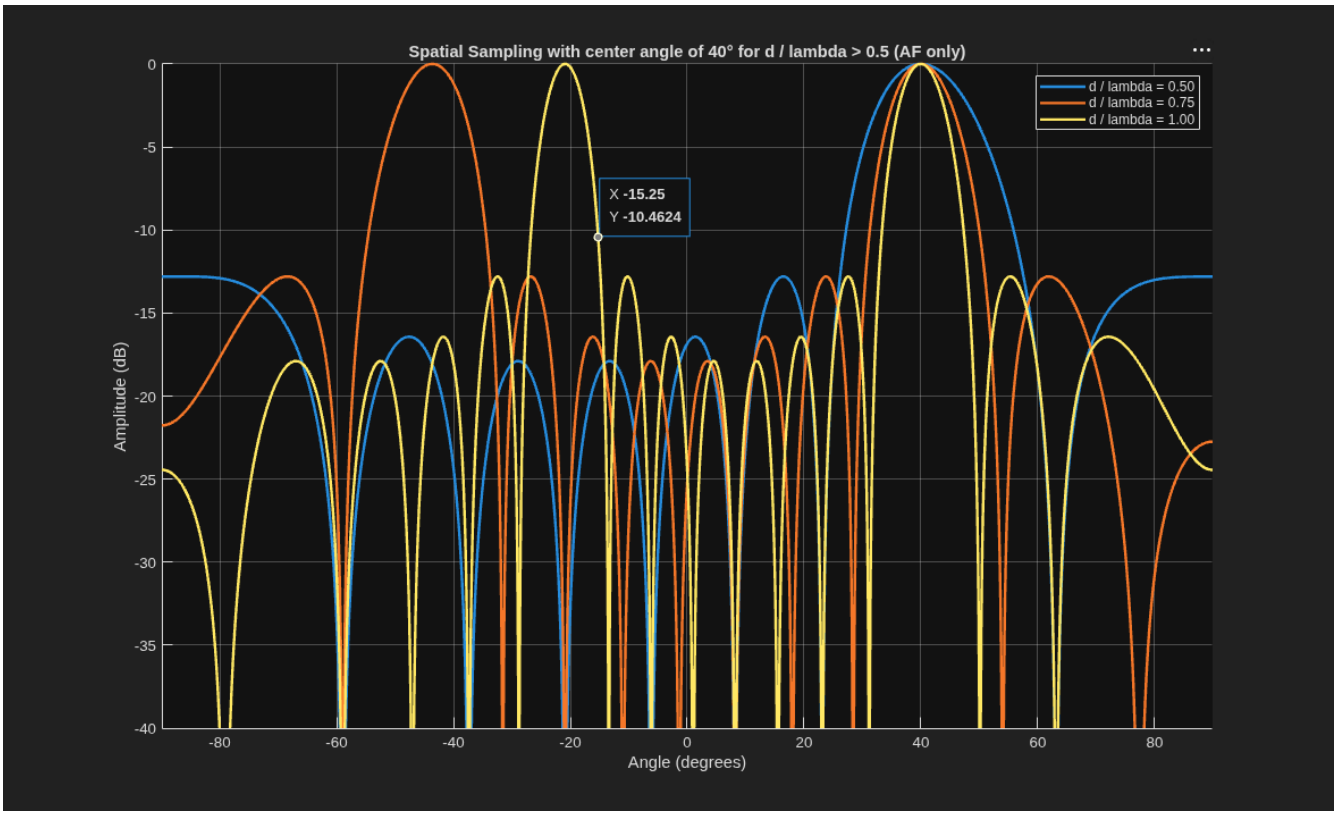


Figure 4: Spatial Sampling for different distance / wavelength ratios ≥ 0.5

In contrast, the ratios of 0.1 and 0.25 produce extremely broad peaks and higher HPBWs. The result shows the basic trade-off between electrical aperture and spatial aliasing: reducing d/lambda broadens the beam, while excessively increasing it creates grating lobes.

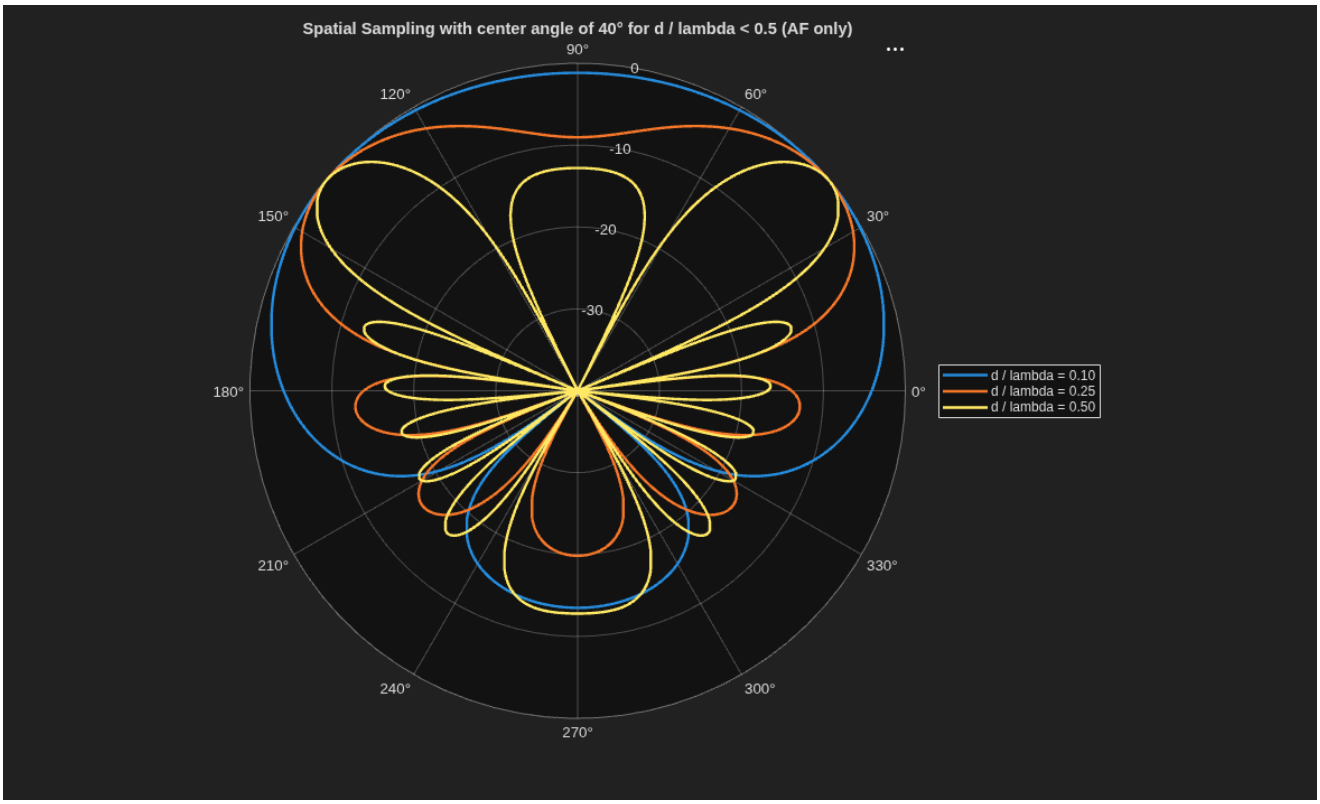


Figure 5: Spatial Sampling for different distance / wavelength ratios ≤ 0.5

Amplitude Weighting (weighting.m)

In this file, different amplitude-weighting algorithms are tested, specifically Hamming and Taylor weighting, with uniform weighting used as a baseline. For each case, the HPBW and Peak Side-Lobe Level (PSLL) are calculated. From both the Cartesian and polar plots and the numerical results, Hamming weighting produced the strongest sidelobe suppression in the tested case, whereas uniform weighting produced the lowest HPBW and therefore the narrowest main beam. The expected trade-off is that stronger sidelobe suppression generally comes at the cost of a wider main beam. [4]

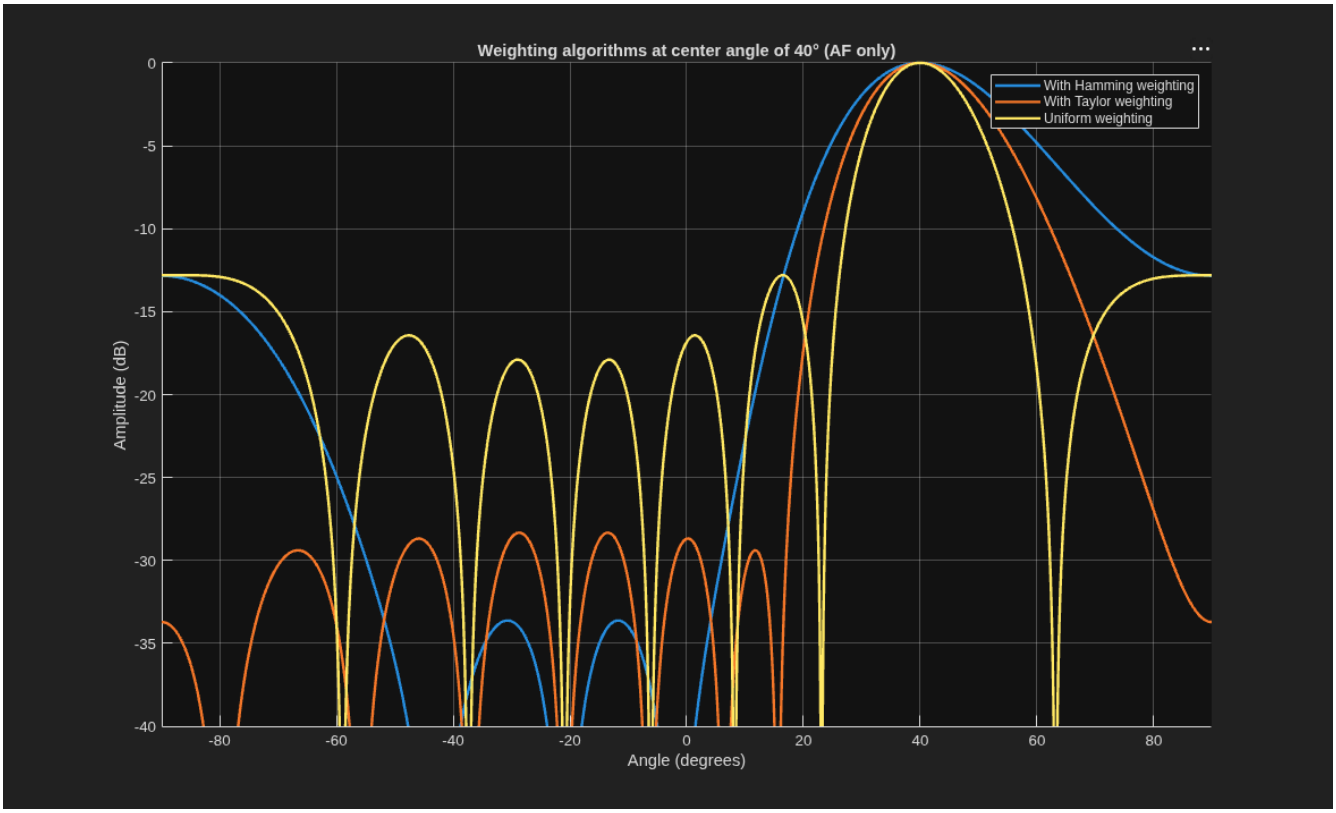


Figure 6: Weighting algorithms with steering angle of 40°

Multibeam Beamforming (multibeam.m)

In the last file of the phase shifter implementation, two beams at -25 and +35 degrees are formed as a representative multibeam case and shown in a Cartesian plot. Numerically, the extracted beam peaks have an error of about 0.1°. This small residual shift is produced by the superposition of the two beamforming weight distributions, while the two normalized peak levels are equal to numerical precision.

Next, operating-frequency ratios f/f_0 of 0.8, 1 and 1.2 are simulated, and the beam-angle errors are calculated for each case. The smallest error occurs at the target frequency, while the largest is approximately 10° for the +35° target beam at $f/f_0 = 0.8$, where the peak moves to around +45°. This shows that each beam in a phase-shifter multibeam solution is still subject to beam squint.

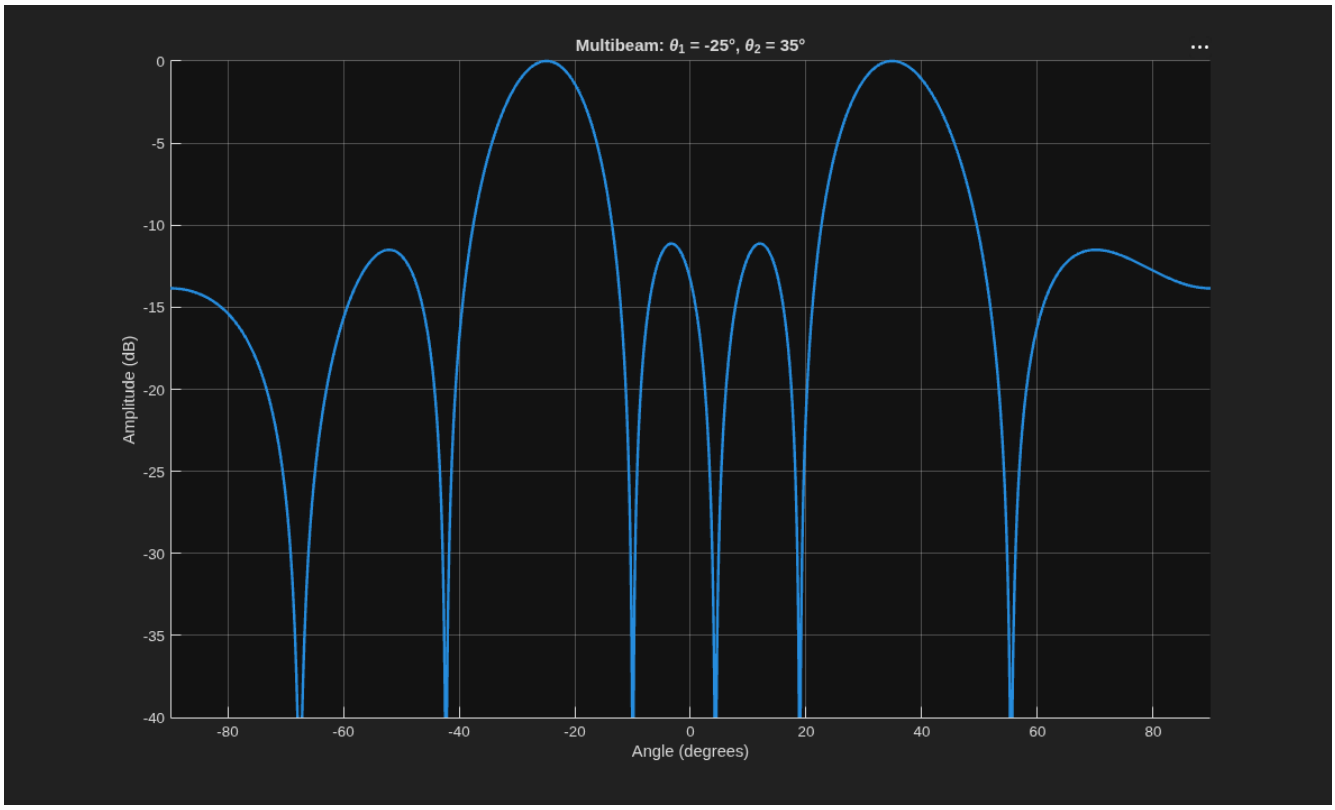


Figure 7: Multibeam Beamforming for 2 target angles

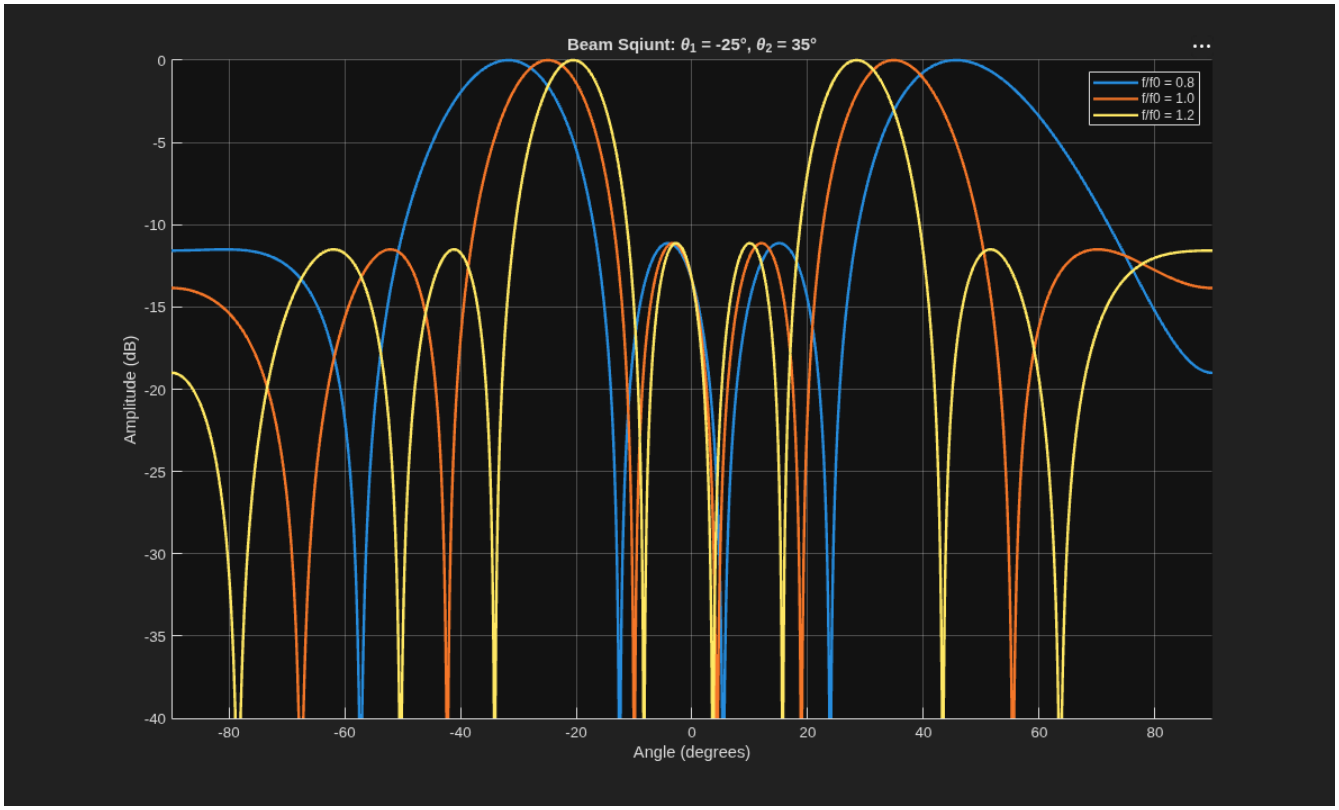


Figure 8: Beam Squint for multibeam beamforming

True-Time-Delay Beamforming (TTD.m)

This file explores the True-Time-Delay architecture. Instead of fixing a phase progression at f_0, a physical time delay is applied between adjacent elements. The required delay step is

$$\Delta\tau = d \cdot \sin(\theta_0) / c$$

$$AF_{TTD}(\theta,f) = \sin\{N \cdot \pi \cdot [d / \lambda(f)] [\sin(\theta) - \sin(\theta_0)]\} / \{N \cdot \sin\{\pi \cdot [d / \lambda(f)] [\sin(\theta) - \sin(\theta_0)]\}\}$$

Just as in the phase shifter implementation, the beamforming results are plotted in Cartesian and polar coordinates for 0, 20 and 40 degrees, together with the time delay required for each target angle. Different frequencies are then analyzed. The TTD architecture does not exhibit beam squint because each element phase follows phi(f) = -2 pi f tau, so the relative delay remains constant with frequency. The beamwidth and sidelobe positions can still change with frequency because the electrical aperture d/lambda changes.

At the end, the pointing errors of the PS and ideal TTD implementations are compared for a 30° steering angle. The phase-shifter error varies across frequency, while the ideal TTD peak remains at the target angle to numerical precision.

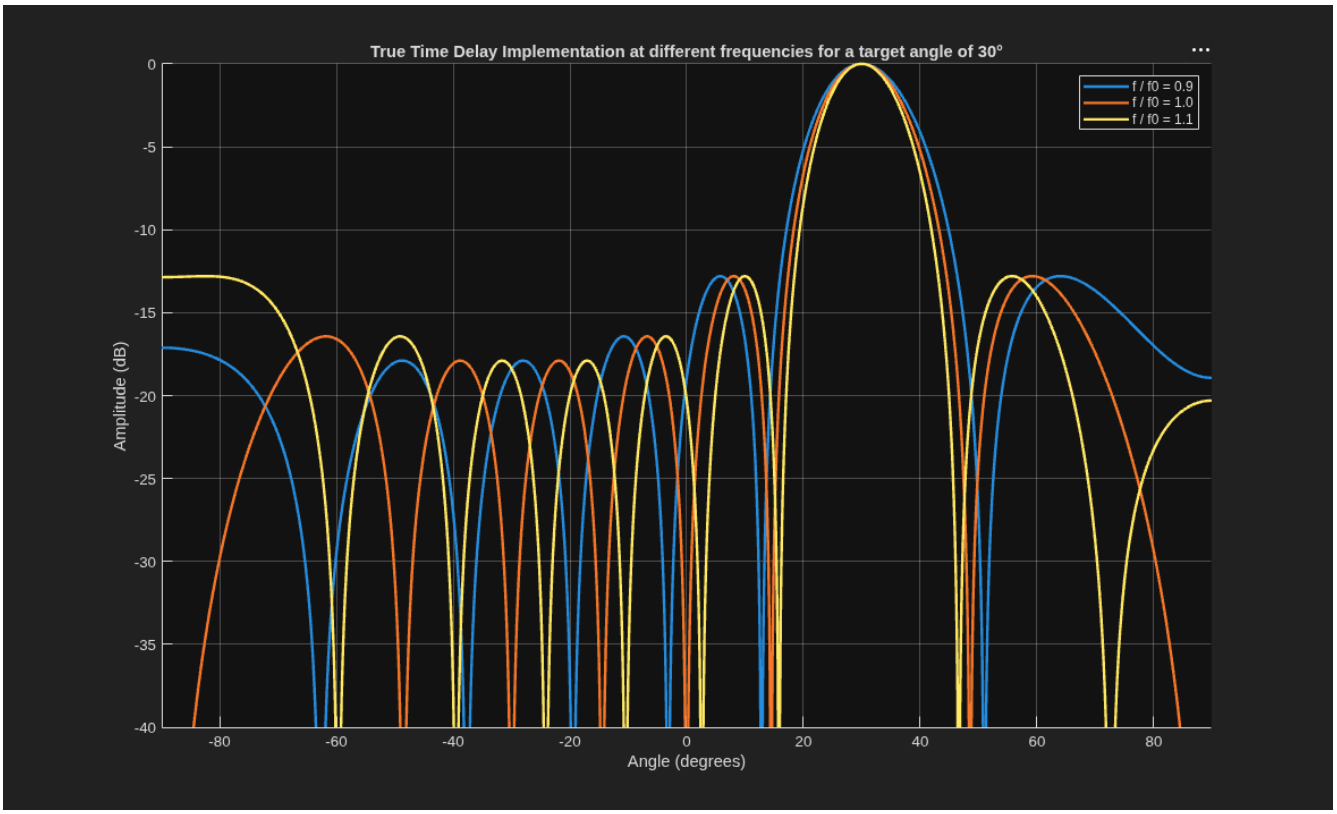


Figure 9: Lack of Beam Squint for true-time-delay architecture

Wideband PS vs TTD Sweeps (worst_cases.m)

This file performs two wideband sweeps over fractional bandwidths from 5 to 50% and steering angles from 0 to 60 degrees, one for each architecture. The purpose is to move from individual examples to a map of where each beamforming architecture begins to fail.

For the TTD architecture, the worst-case sidelobe level is calculated. The sidelobes remain approximately constant at around -12.7 dB over most of the sweep. At sufficiently high frequency and large steering angle, however, a grating lobe enters the visible region and can reach the same strength as the primary beam. For the fixed physical spacing $d = \lambda_0/2$, the electrical spacing is $d/\lambda(f) = f/(2f_0)$, so the grating-lobe risk increases toward the high-frequency edge of the band, not the low-frequency edge.

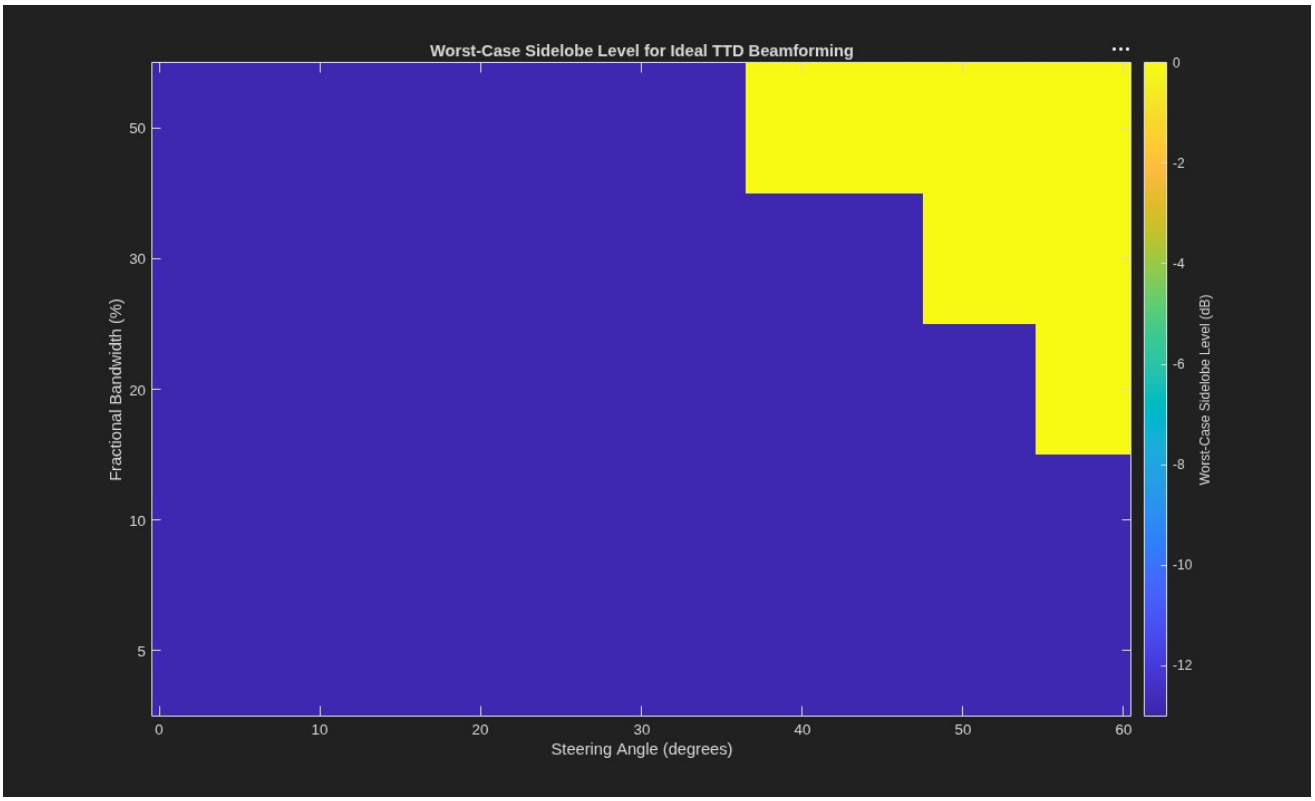


Figure 10: Worst-case TTD sidelobe level vs fractional bandwidth and steering angle

For the PS architecture, the same sweep is performed for the worst-case pointing error. The increase is gradual as bandwidth and steering angle increase. When $|(f_0/f) \cdot \sin(\theta_0)|$ exceeds 1, however, the closed-form arcsin relation has no real visible steering solution.

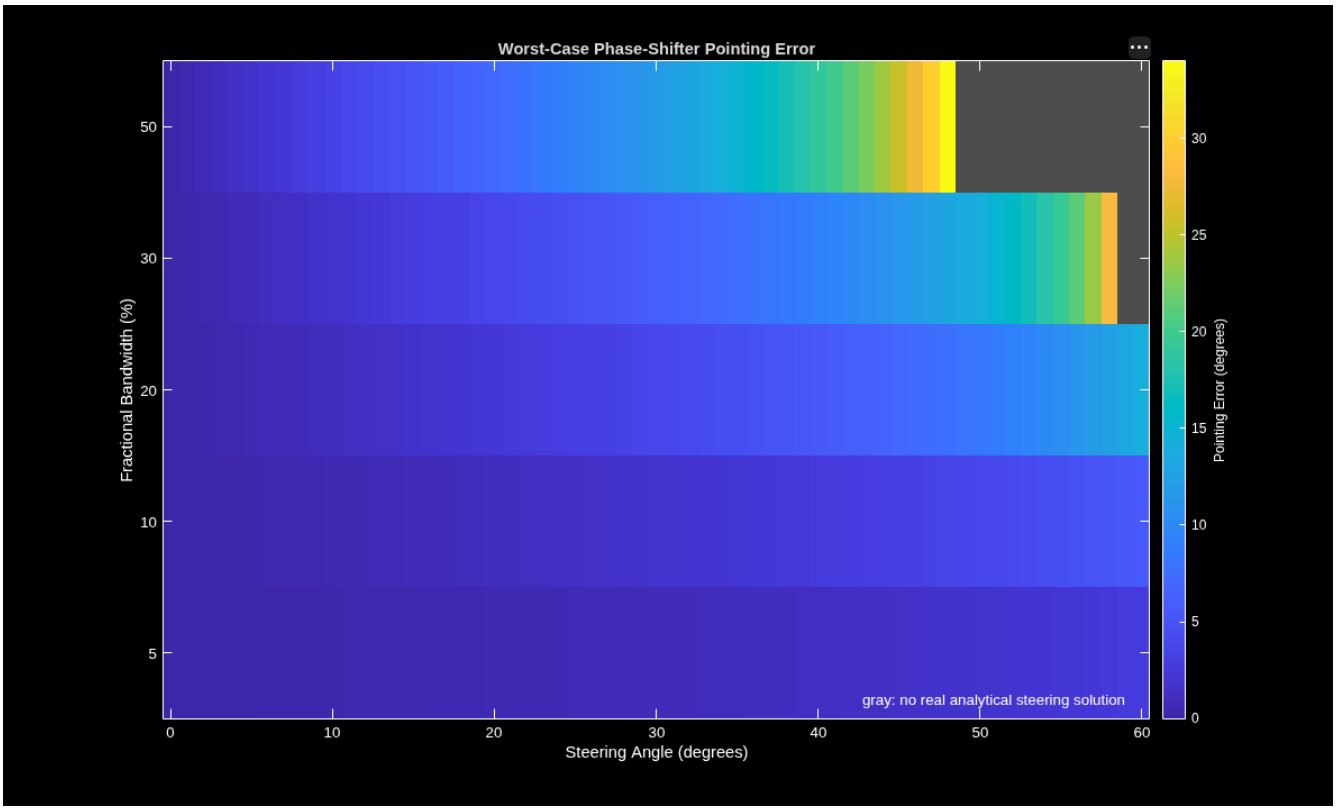


Figure 11: Worst-case PS pointing error vs fractional bandwidth and steering angle

RF Non-Idealities (RF_nonidealities.m)

The last part of this section includes an examination of non-idealities that exist in real-world beamforming systems. 2-, 4- and 6-bit phase shifter implementations are calculated, with a phase step of 360° / 2^B for B bits. Their quantization error and peak angle are calculated and the resulting patterns are plotted against the ideal beam.

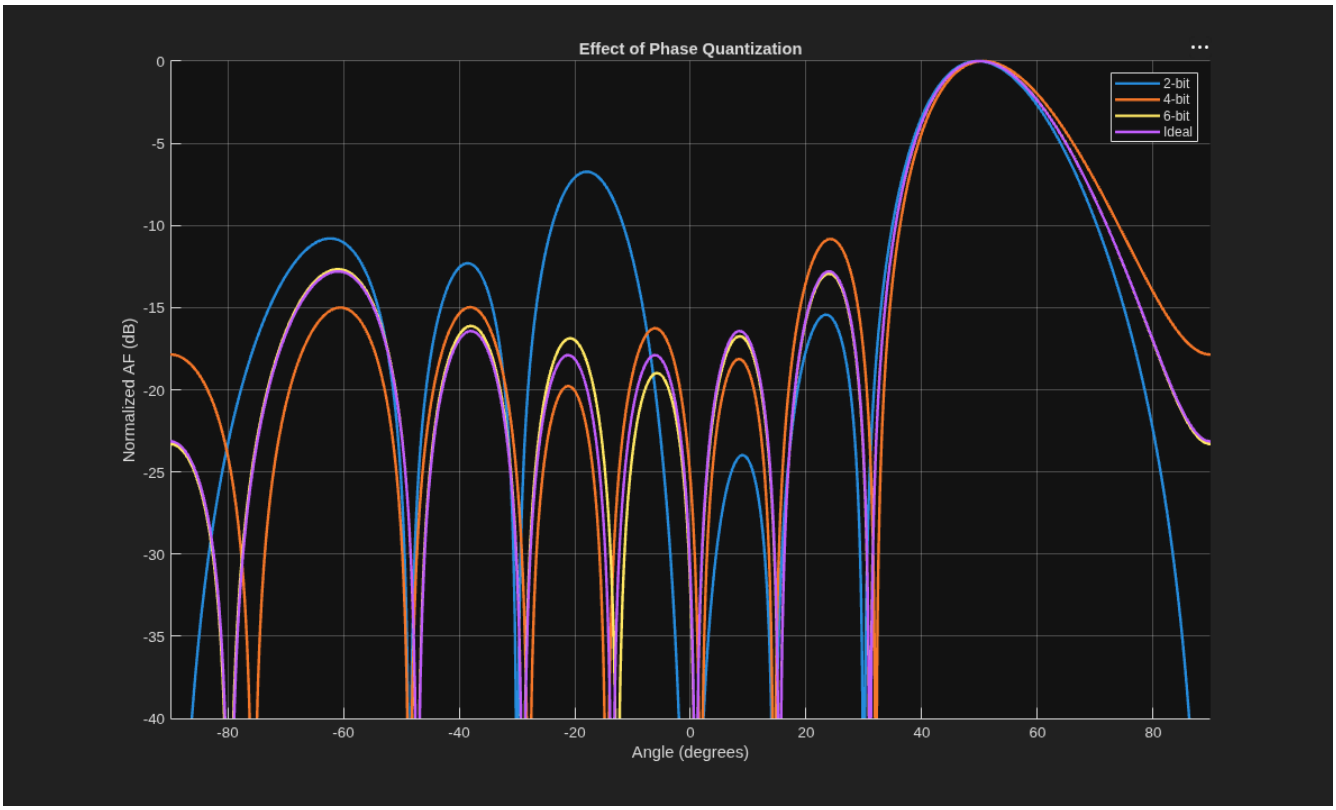


Figure 12: 2-, 4-, and 6-bit phase shifter implementation

Then, for the PS architecture, a Monte Carlo analysis is performed with 5000 trials. For each trial and each element, independent Gaussian amplitude and phase perturbations are regenerated with standard deviations $\sigma_A = 0.2$ in relative amplitude and $\sigma_\varphi = 20°$. The pointing error distribution is calculated, and a representative Cartesian plot of a beam containing non-idealities is also shown against the ideal one.

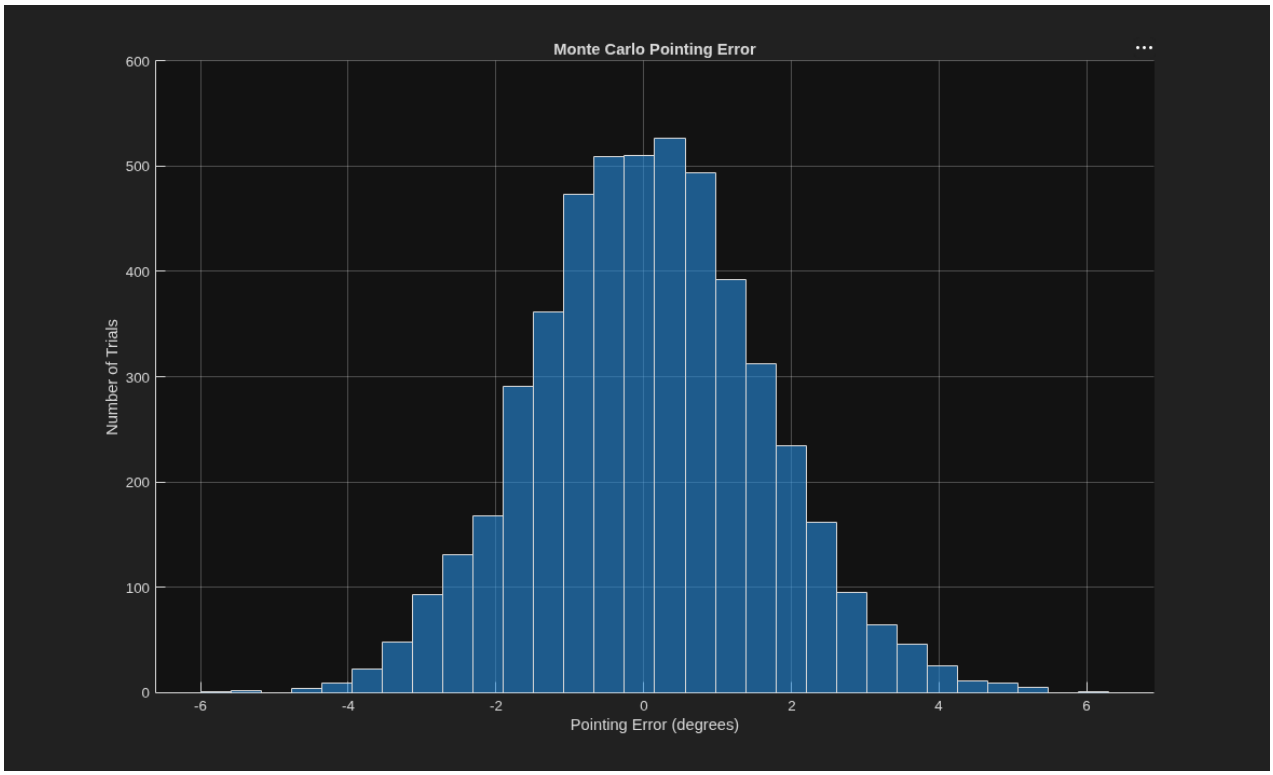


Figure 13: Monte Carlo Analysis of pointing error for random phase & amplitude mismatch

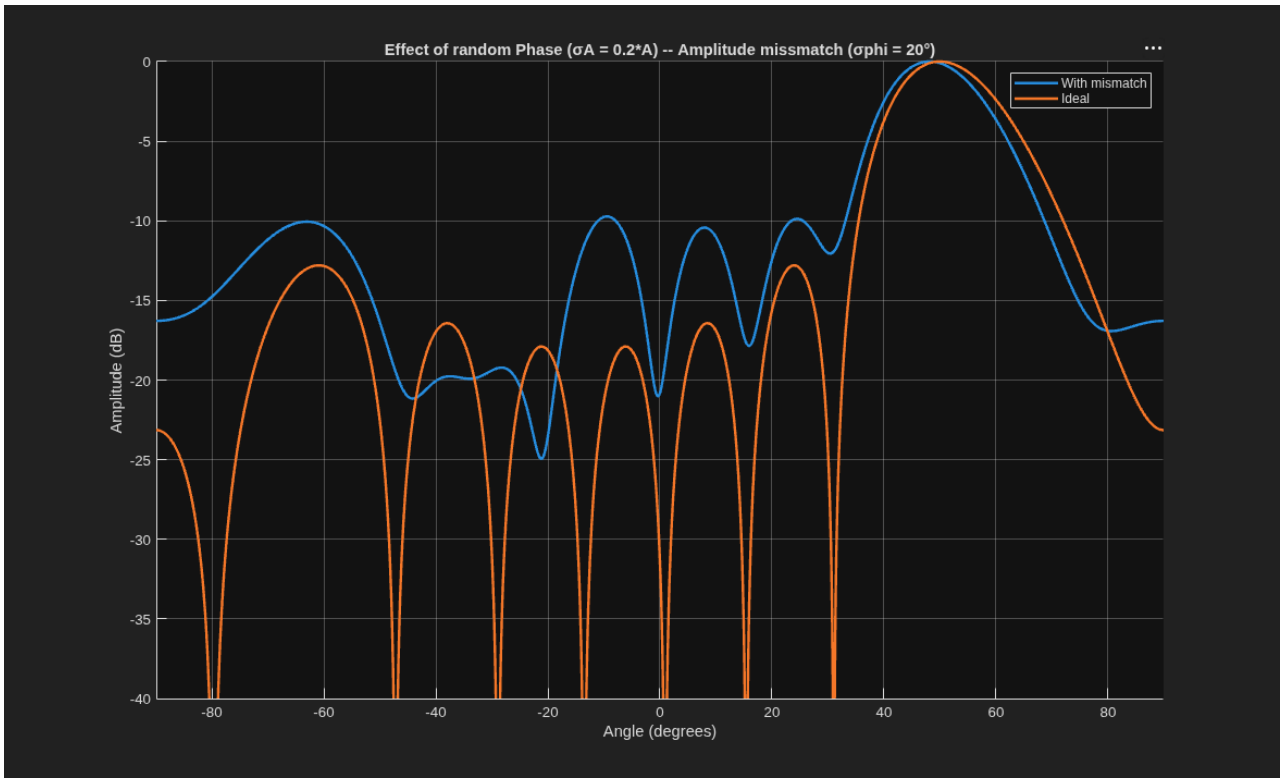


Figure 14: Beam with random phase & amplitude mismatch against ideal

Lastly, for the TTD architecture, a similar Monte Carlo analysis is performed with the same amplitude and phase mismatch. For every trial, the main lobe is excluded before the highest remaining local maximum is recorded, so the resulting histogram represents the distribution of the Peak Side-Lobe Level rather than accidentally selecting the main beam.

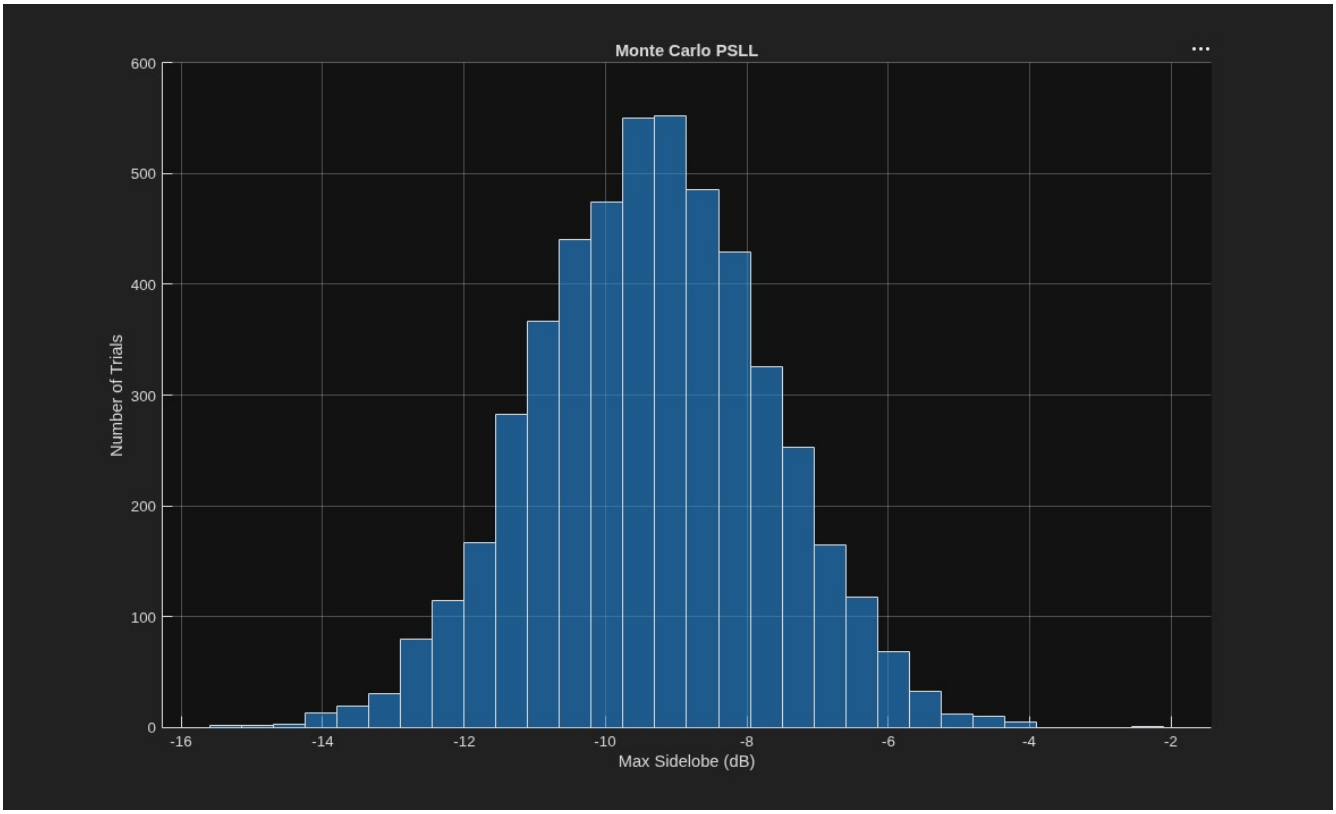


Figure 15: Monte Carlo Analysis of PSLL for random phase & amplitude mismatch

At this point in the project, the majority of the phased-array analysis and calculations had been completed, so the focus turned to implementing in ANSYS INTERCONNECT the TTD architecture that had so far only been analyzed theoretically.

PART 3: ANSYS INTERCONNECT PHOTONIC TTD

Ideal Optical Model (set_ideal.lsf)

This script contains the parameters for an ideal optical TTD implementation. All loss and noise effects are disabled, while the Mach-Zehnder modulator (MZM) presents an extinction ratio of 100 dB. This case is used as an ideal reference rather than as a realistic component model. [5]-[8]

Non-Ideal Optical Model (set_non-ideal.lsf)

This script contains the parameters for a non-ideal TTD implementation. The MZM element has an insertion loss of 6 dB and an extinction ratio of 30 dB. Each WGD receives a random channel-

dependent length perturbation bounded by ±0.5 mm relative to the nominal design, together with a propagation loss of 200 dB/m and a dispersion coefficient of 2.8e-3 s/m². [6], [7]

Thermal + Noise Case (set_thermal.lsf)

This script contains a separate thermal + noise stress case. The temperature is increased to 320 K from the 300 K nominal temperature. The waveguide (WGD) has thermal effects enabled, with an effective index temperature sensitivity of 2e-4 K^{-1} and a thermal fill factor of 1. The group-index temperature coefficient is dn_g/dT=5e-4 K^{-1}. The corresponding temperature-dependent group index estimate is calculated as

$$n_g(T) = n_g(T_0) + (dn_g / dT) \cdot (T - T_0)$$

Thermal noise for the photodetector elements is also enabled with a magnitude of 1e-13 $A/\sqrt{Hz}$. The extracted response shows channel-dependent phase fluctuations and delay errors of up to approximately 1.9 ps.

For all three cases, the nominal group index n_g and effective index n_{eff} are set to 4.2 and 2.39 respectively.

Photonic TTD Network (beamforming_project.icp)

This file contains the actual implementation of the optical True-Time-Delay network. The Network Analyzer (ENA) element in INTERCONNECT is used to characterize the electrical transfer response of all eight channels over the RF frequency sweep around the 10 GHz center frequency.

Firstly, a continuous-wave optical signal is produced by the Continuous Wave Laser (CWL) element of INTERCONNECT at 193.1 THz and a power of 0 dBm. The optical carrier is modulated by the RF signal using an MZM with a bias voltage of 2 V, and the optical output is split into 8 branches, one for each array element. [5], [6]

Afterwards, each optical signal is sent through a waveguide of a different length in order to produce the required time delay. The delay-length relation used for the system-level model is

$$\tau = n_g \cdot L / c$$

$$\Delta L = c \cdot \Delta \tau / n_g$$

For the 30° steering case with d = $\lambda_0/2$, the adjacent-element delay is 25 ps. With n_g = 4.2, this corresponds to an adjacent waveguide-length difference of approximately 1.79 mm. Finally, each delayed optical signal is detected by a PIN photodetector and returned to the ENA as an electrical signal. Noise is disabled in the ideal and randomized non-ideal cases so that their complex transfer functions and delays can be extracted cleanly; detector thermal noise is enabled only in the separate thermal + noise stress case. [7], [8]

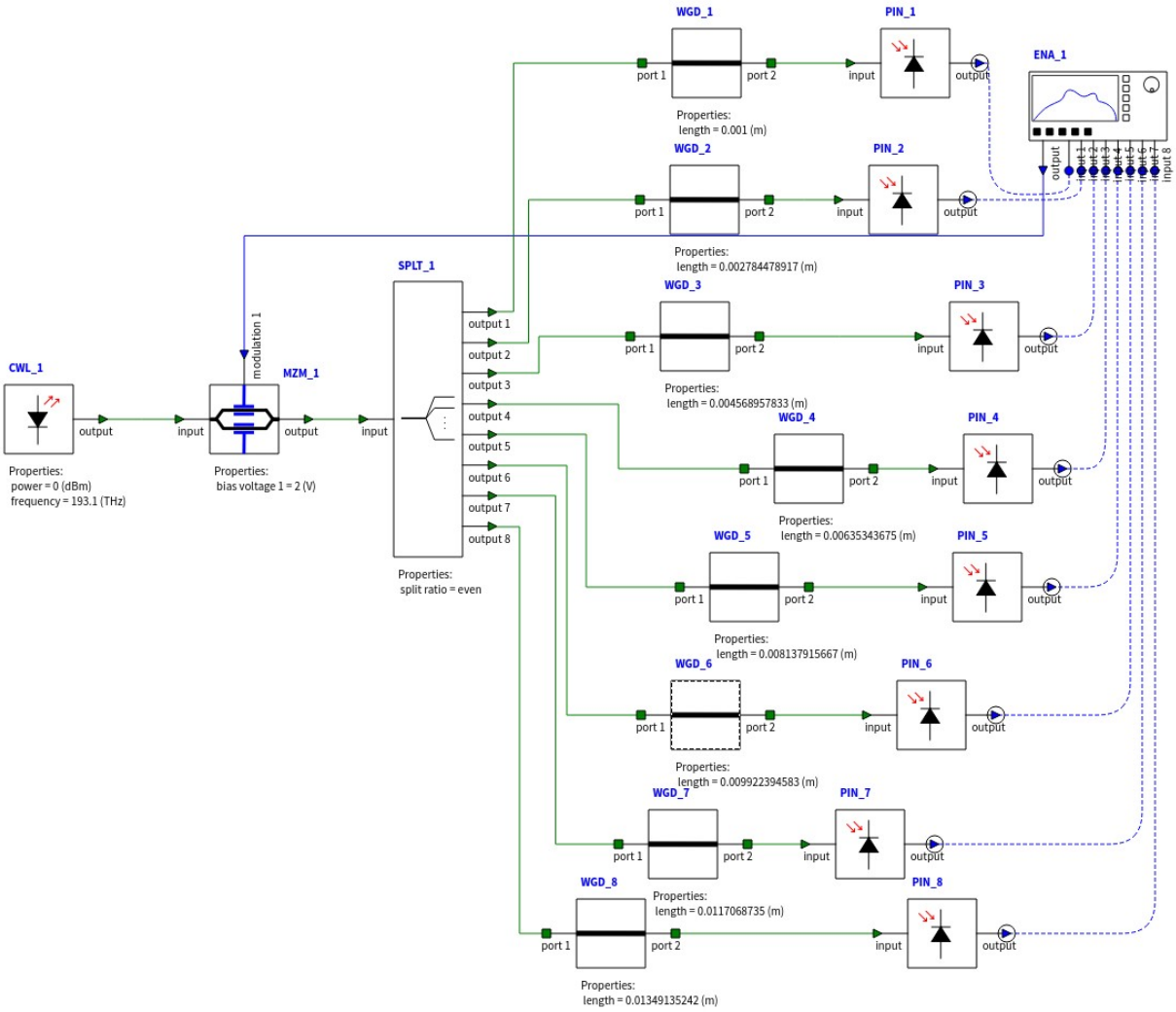


Figure 16: Optical True-Time-Delay Implementation

Response Extraction and Validation (extract_export.lsf)

This script contains the complete analysis of the INTERCONNECT simulation. Firstly, the absolute complex transfer functions $H_n(f)$ and the relative responses

$$H_{rel,n(f)} = H_1(f) / H_n(f)$$

are collected. The relative phase is unwrapped and the differential delay is extracted from its slope using

$$\tau = -(1/2\pi) \cdot d\varphi/df.$$

The plotted graphs compare required and measured delay, delay error and relative phase, while the absolute response is also retained so that insertion loss is not hidden by normalization. Finally, the information gathered from the optical simulation is exported to MATLAB for the end-to-end

comparison. In the ideal implementation, the measured delays follow the theoretical values to numerical precision. The randomized non-ideal realization and the thermal + noise case introduce separate channel-dependent magnitude, phase and delay deviations.

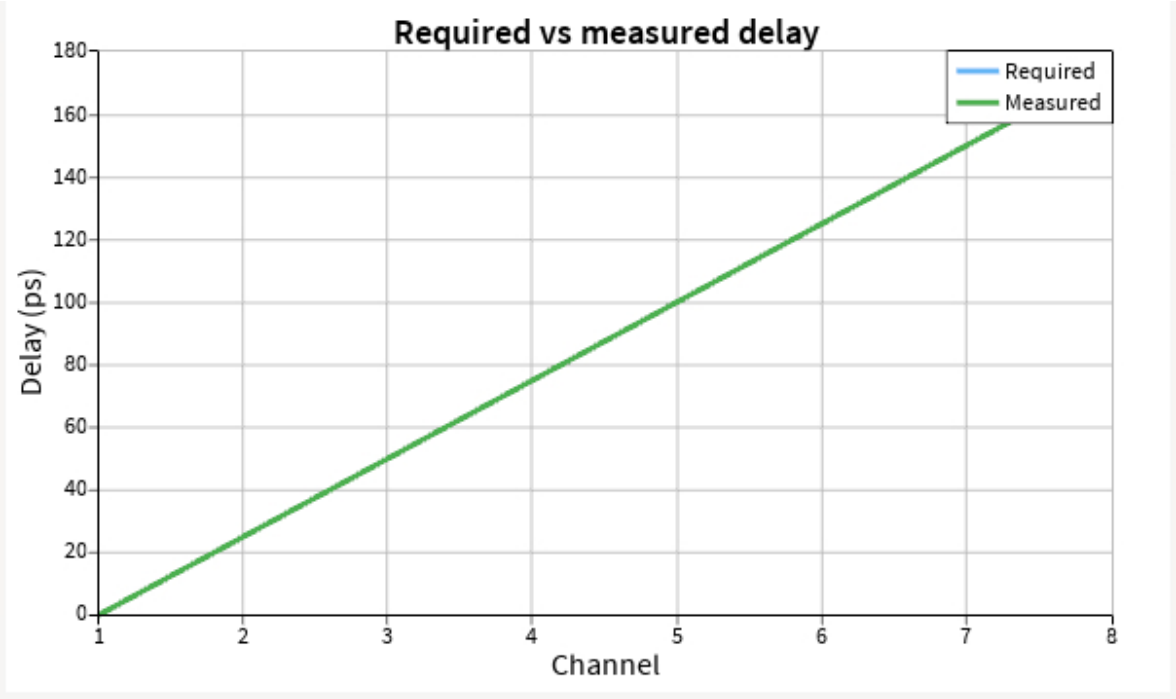


Figure 17: Required vs Measured Delay of ideal optical implementation

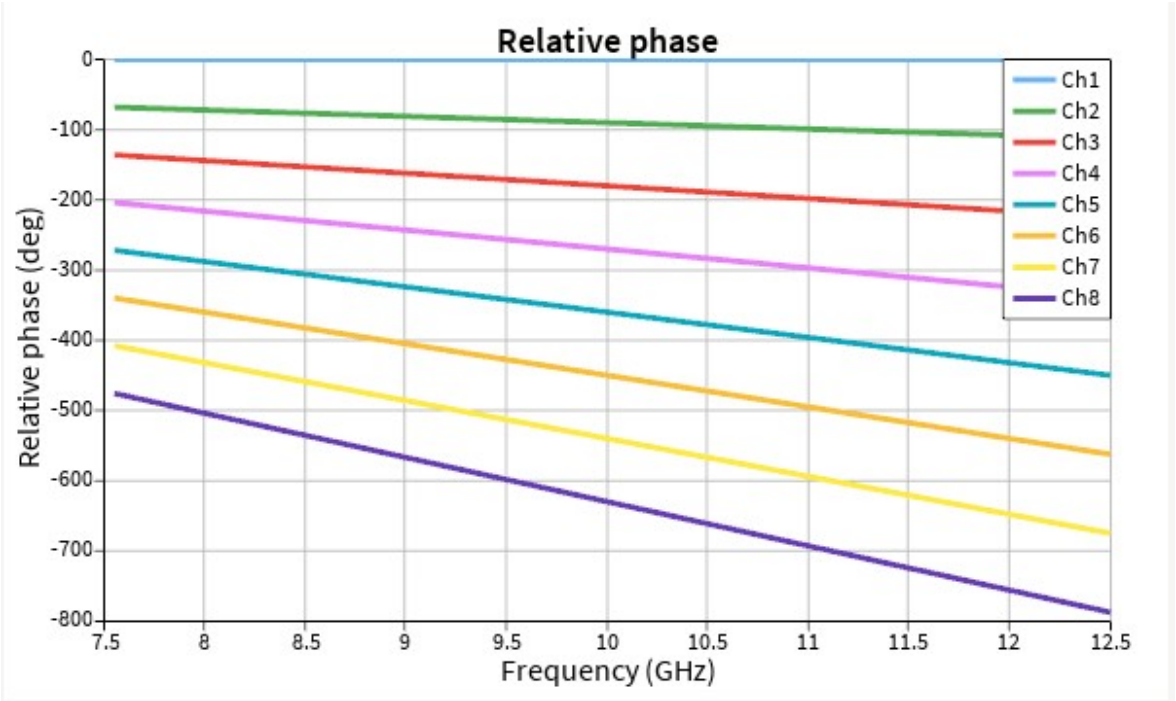


Figure 18: Relative phase for each optical channel for ideal implementation

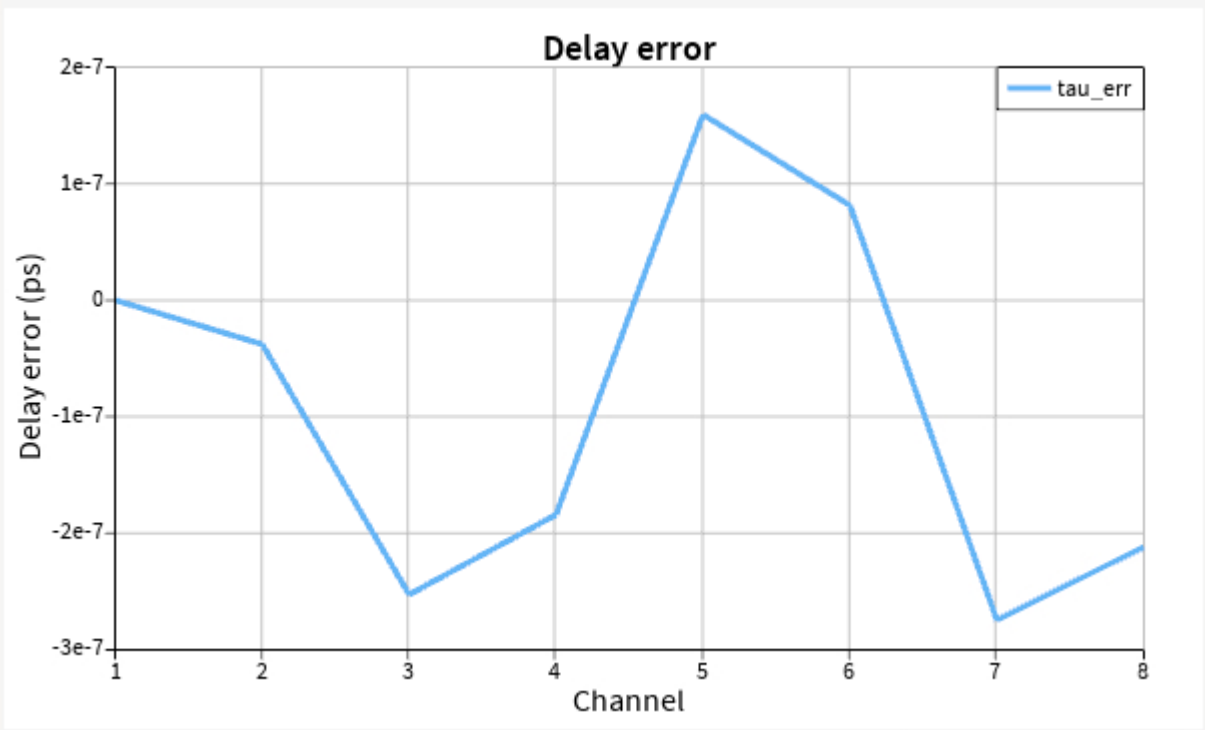


Figure 19: Delay error for ideal implementation



Figure 20: Delay error for non-ideal implementation

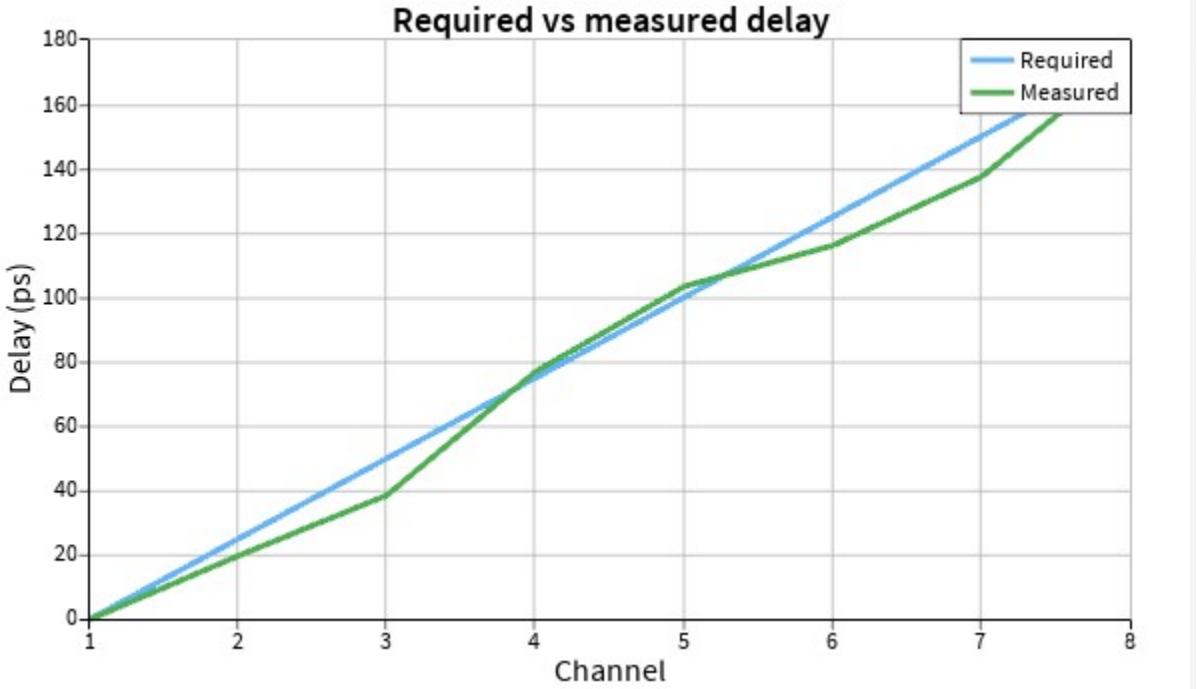


Figure 21: Required vs Measured Delay of non-ideal optical implementation

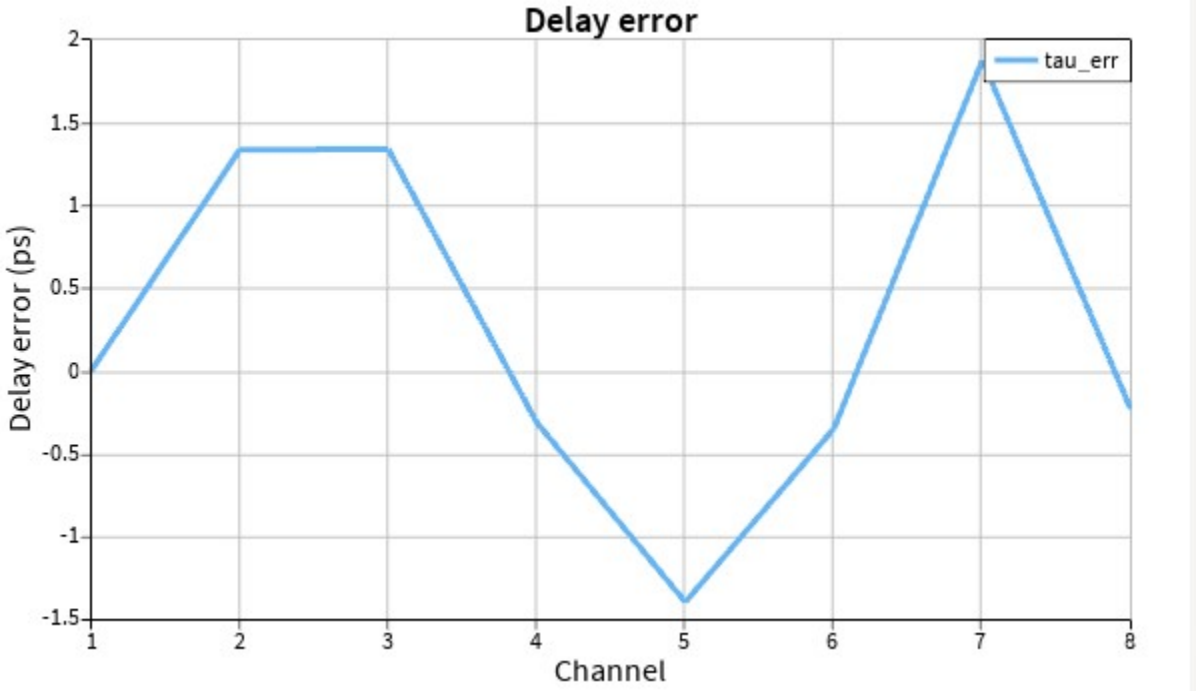


Figure 22: Delay error for thermal + noise implementation

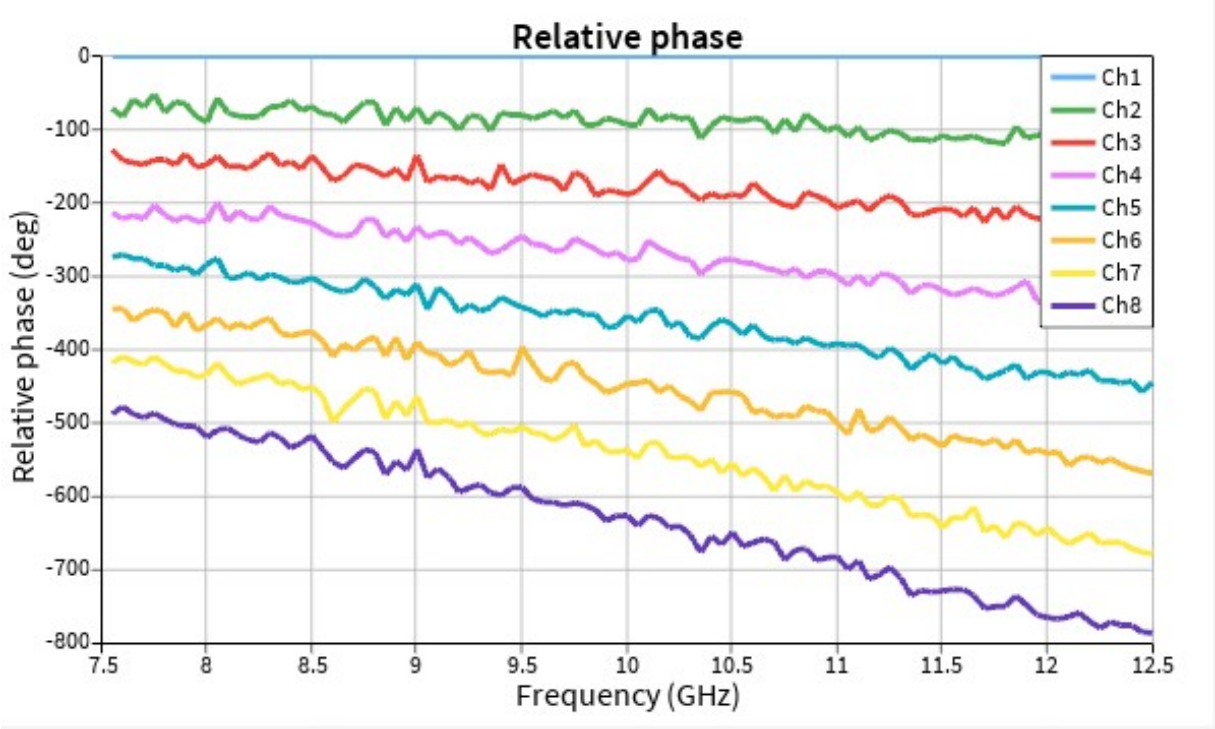


Figure 23: Relative phase for thermal + noise implementation

PART 4: END-TO-END MATLAB - INTERCONNECT COMPARISON

End-to-End Comparison (end_to_end_comparison.m)

The final file of the project compares the fixed phase shifter, ideal True-Time-Delay and simulated INTERCONNECT True-Time-Delay implementations across approximately 50% fractional bandwidth around the 10 GHz center frequency. The INTERCONNECT model is not replaced by an ideal delay in MATLAB; instead, the exported complex response H_n(f) of each photonic channel is applied directly to the array weights. This allows the final beam pattern to contain the phase and magnitude behavior produced by the optical network.

The fixed phase-shifter beam shows the expected systematic beam squint, reaching 11.47° pointing error at the low-frequency edge and crossing zero at the 10 GHz design frequency. The ideal photonic TTD remains at zero error. The non-ideal photonic TTD stays within approximately 0.59° of the ideal case, while the separate thermal + noise case shows larger random deviations with a maximum of approximately 1.31°.

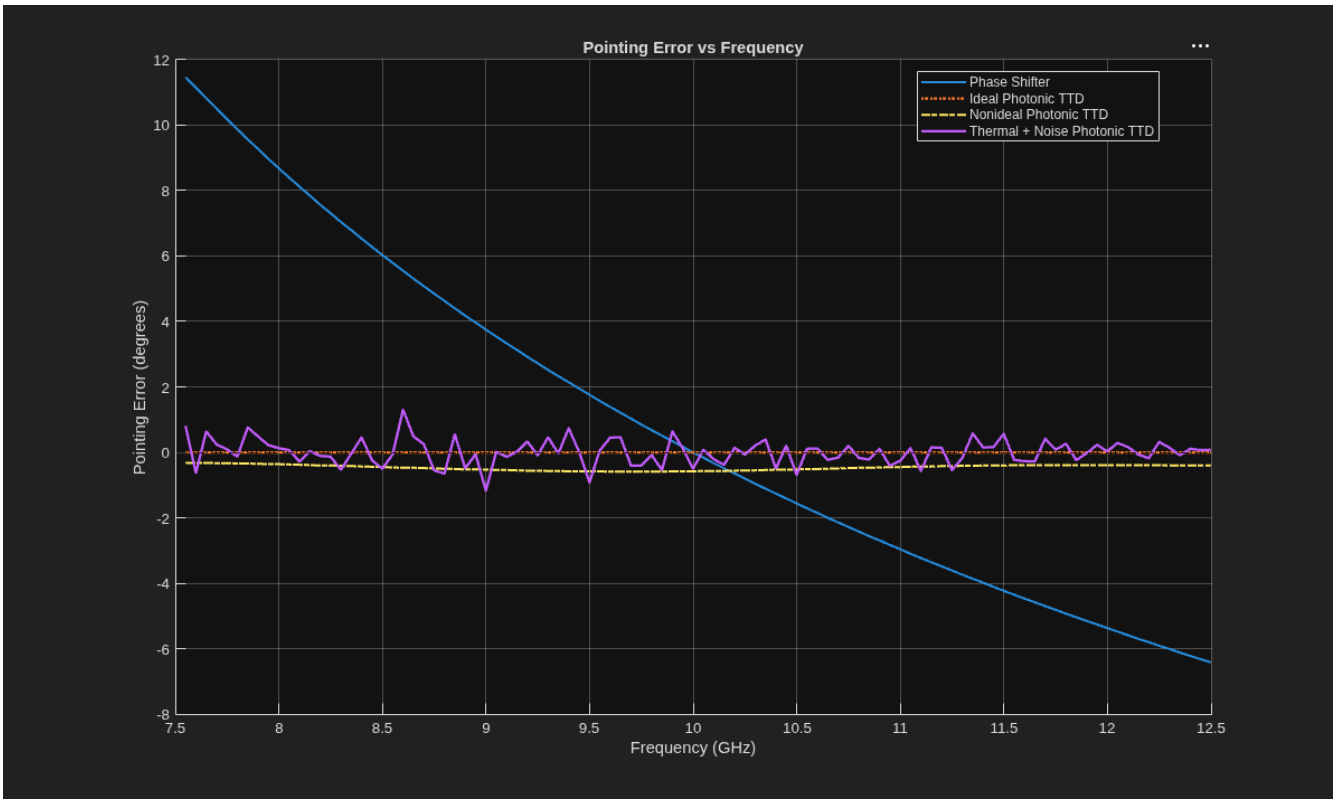


Figure 24: Pointing error vs frequency for the different implementations

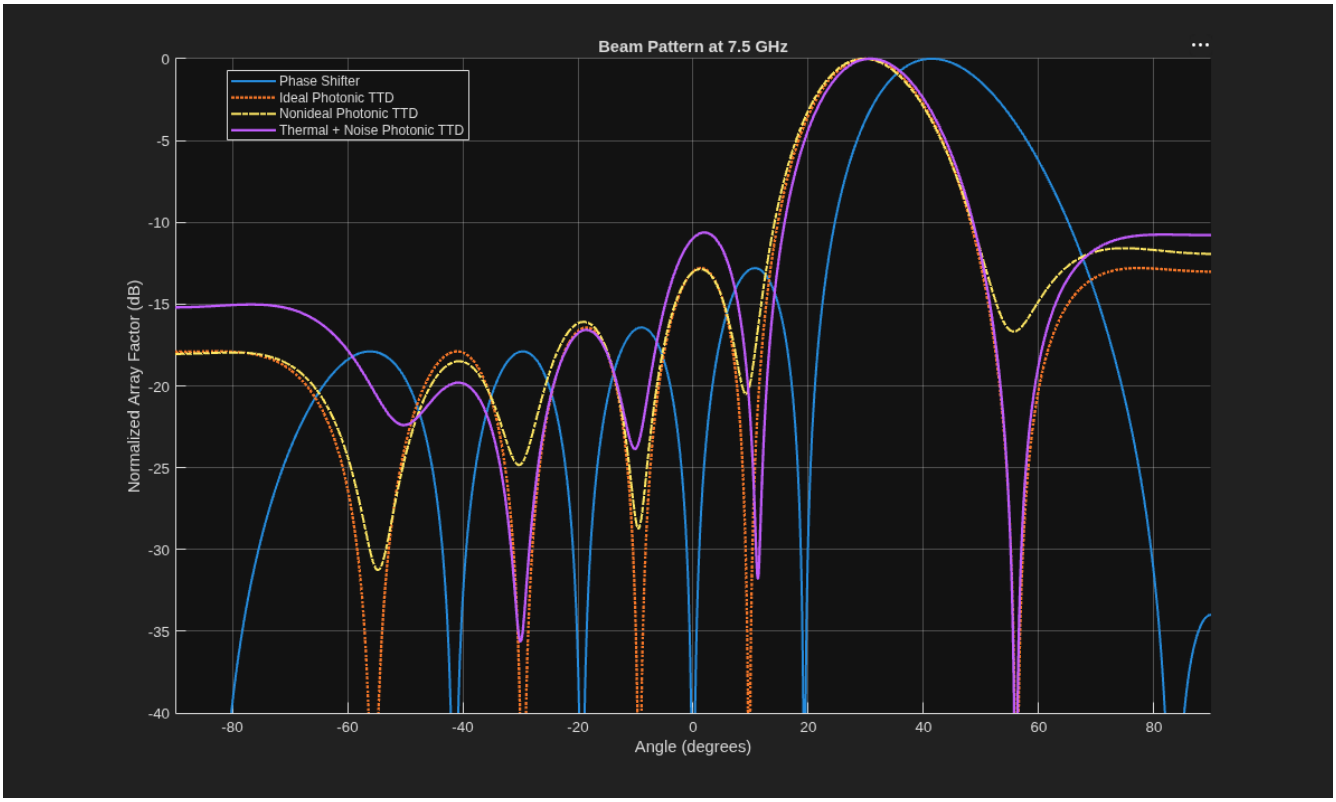


Figure 25: End-to-end comparison for the different implementations at 7.5 GHz

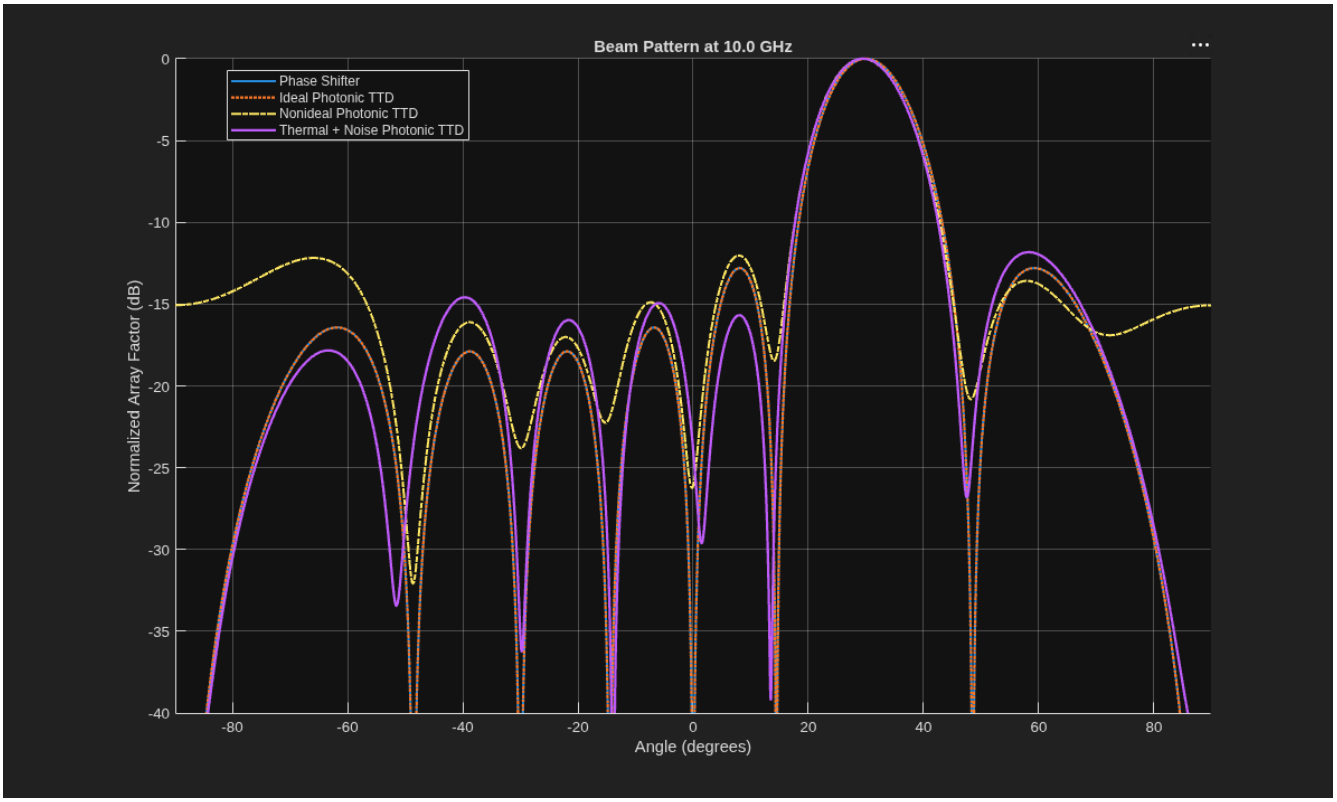


Figure 26: End-to-end comparison for the different implementations at 10 GHz

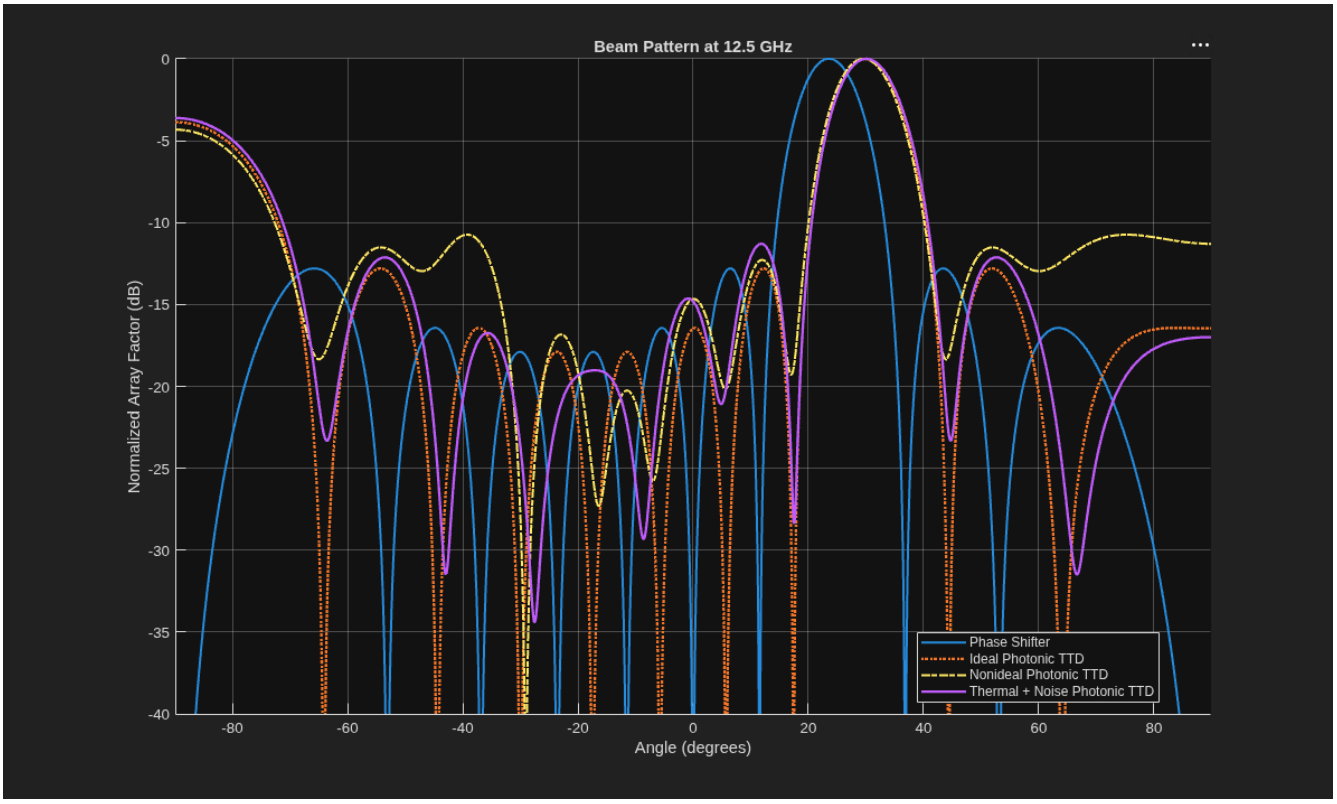


Figure 27: End-to-end comparison for the different implementations at 12.5 GHz

At 10 GHz, all four implementations are centered close to the 30° target. At 7.5 and 12.5 GHz, the phase-shifter peak moves because of beam squint, whereas the ideal, non-ideal and thermal + noise TTD cases remain close to the target direction. The thermal + noise case introduces more pattern variation, but not the systematic frequency-dependent steering error of the fixed phase shifter.

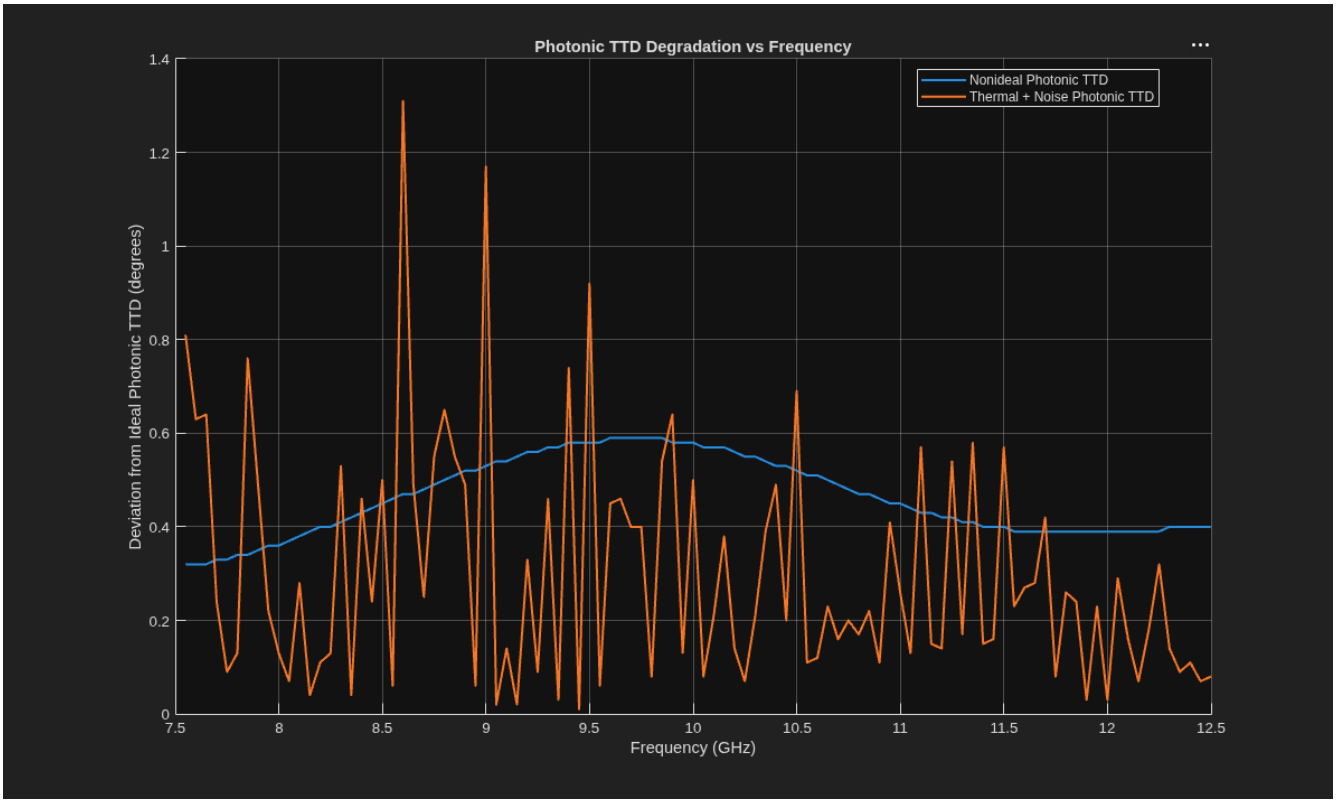


Figure 28: Deviation from ideal TTD of non-ideal & thermal realizations

The direct TTD-to-TTD comparison isolates the photonic degradation from the phase-shifter beam squint. In the current realization, the non-ideal photonic TTD remains within about 0.59° of the ideal TTD over the sweep, whereas the thermal + noise case reaches about 1.31° maximum deviation.

At 10 GHz, the randomized non-ideal case develops a relative-magnitude imbalance that generally increases toward channel 8 and reaches about 4.93 dB by channel 8, while its phase error remains within about 27.3° in magnitude. The thermal + noise case has smaller magnitude variation, approximately -1.7 to +0.9 dB, but more irregular frequency-dependent phase variation/noise, reaching roughly -9 to +6.4 degrees in this realization.

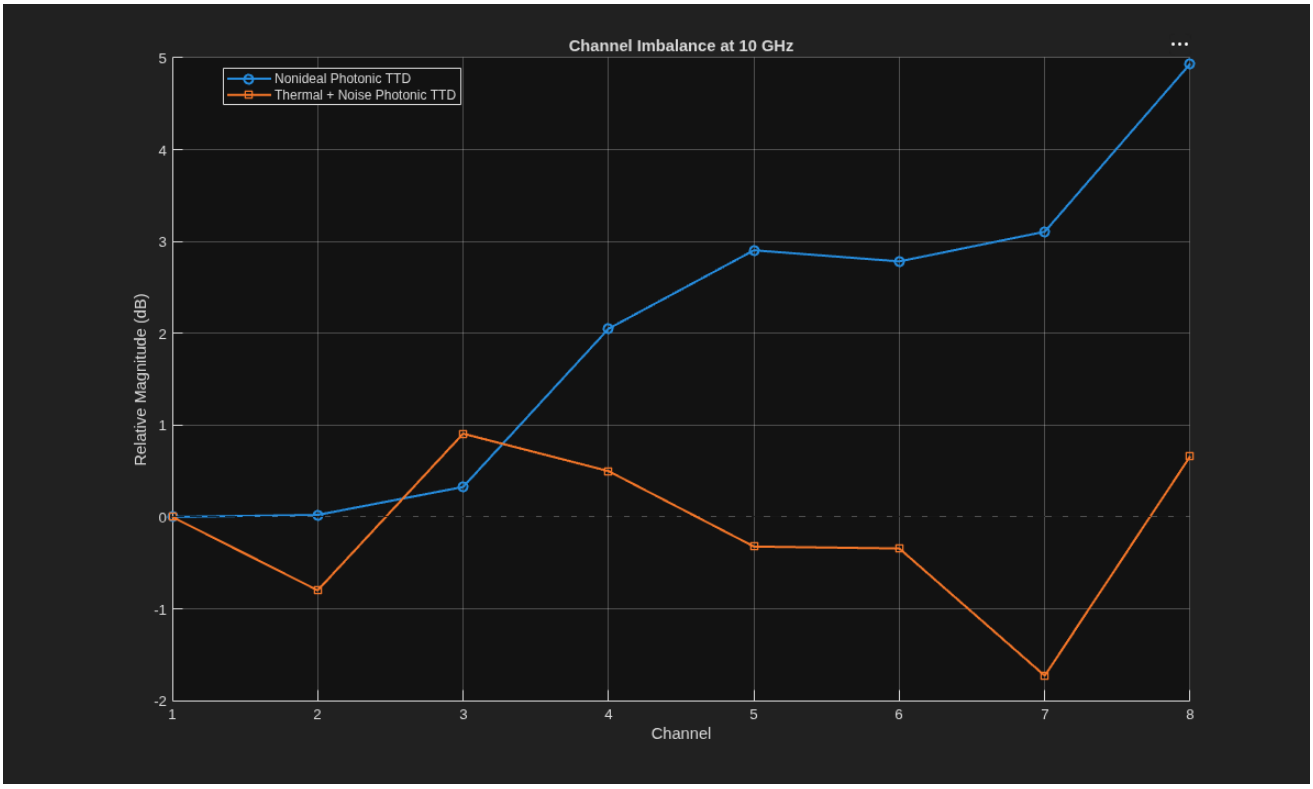


Figure 29: Channel imbalance for non-ideal & thermal realizations

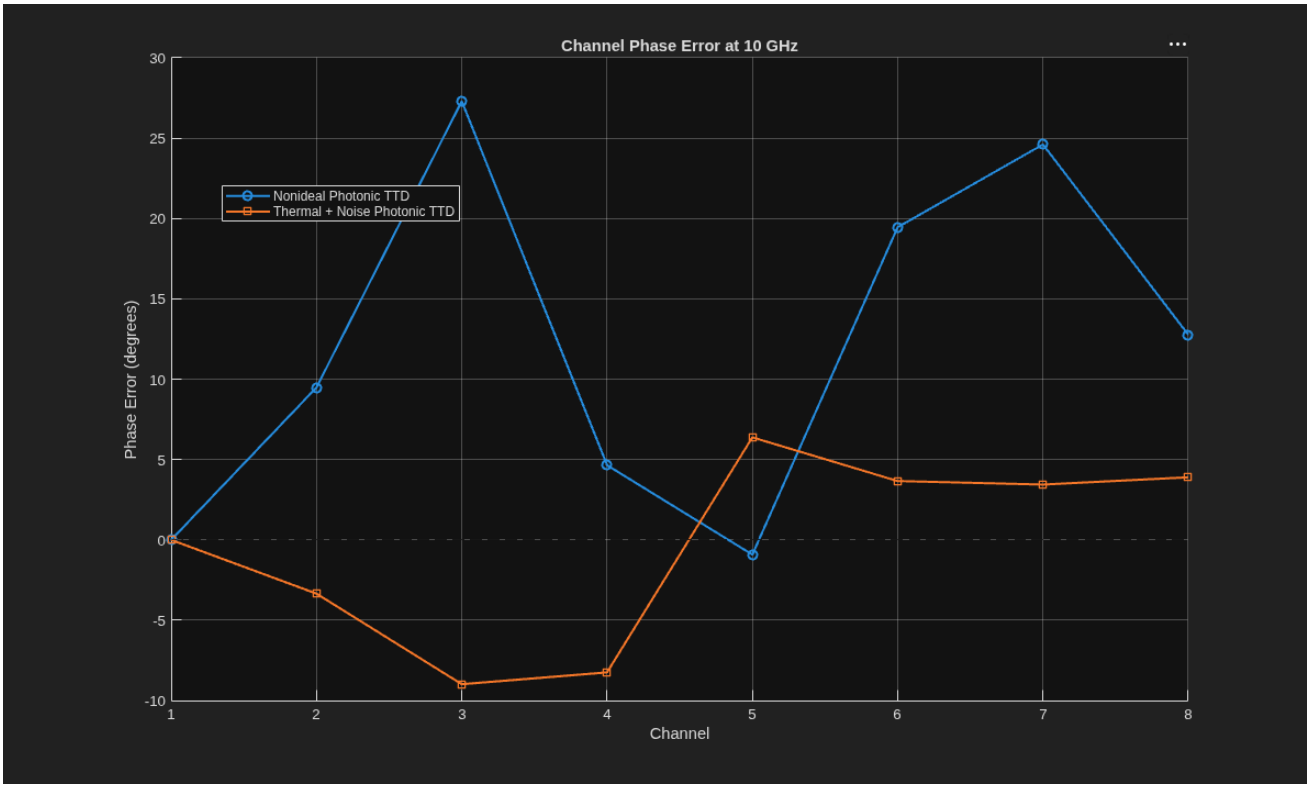


Figure 30: Channel phase error for non-ideal & thermal realizations

The extracted differential-delay error reaches approximately 12.6 ps in magnitude for the randomized non-ideal realization and approximately 1.9 ps for the thermal + noise realization. These are channel-dependent errors rather than common delays, so they directly perturb the relative phase progression used for beam steering.

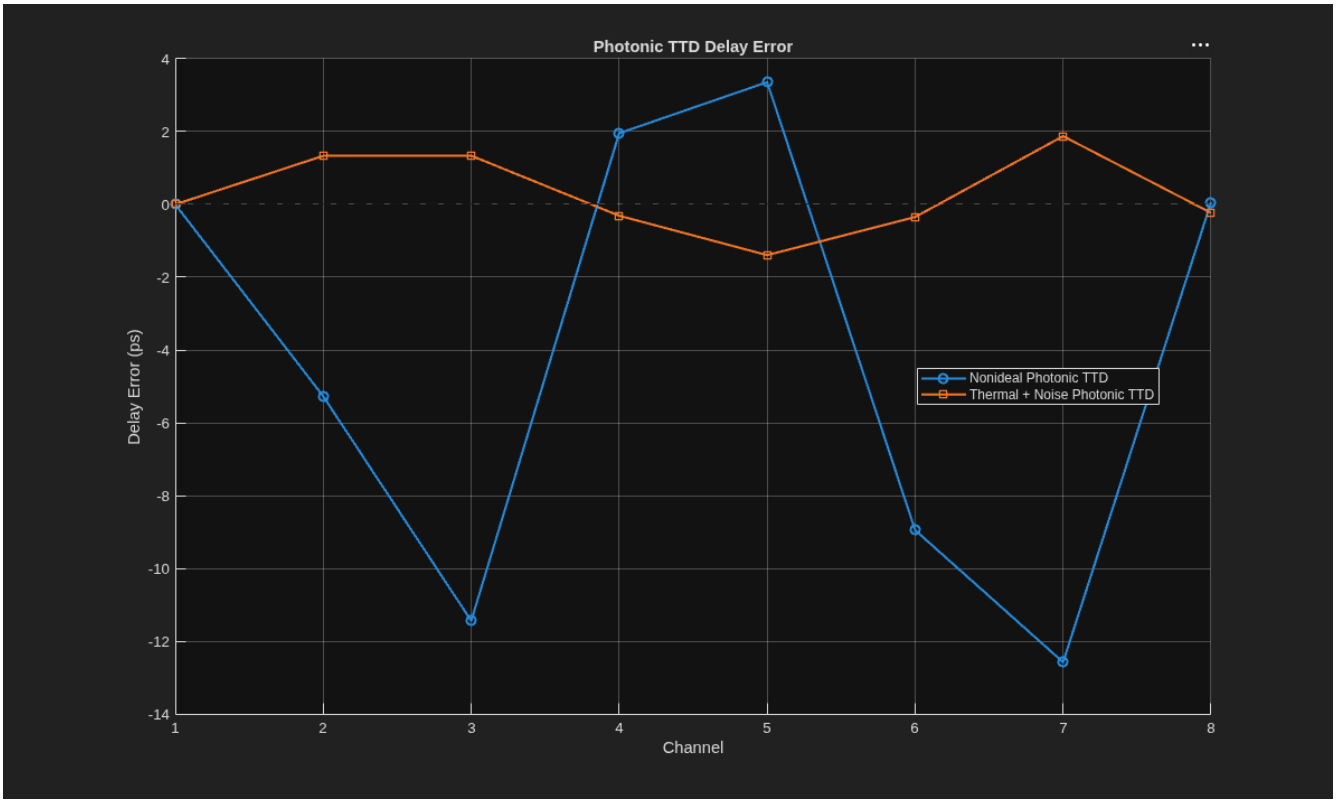


Figure 31: Photonic TTD delay error for non-ideal & thermal realizations

The absolute comparison separates insertion loss from the normalized beam-pattern results. The randomized non-ideal case is approximately 14.8-19.8 dB below the ideal response across the eight channels, whereas the separate thermal + noise case remains within roughly ±1.3 dB of the ideal magnitude. These two cases therefore represent different stress mechanisms and are not cumulative versions of the same model.

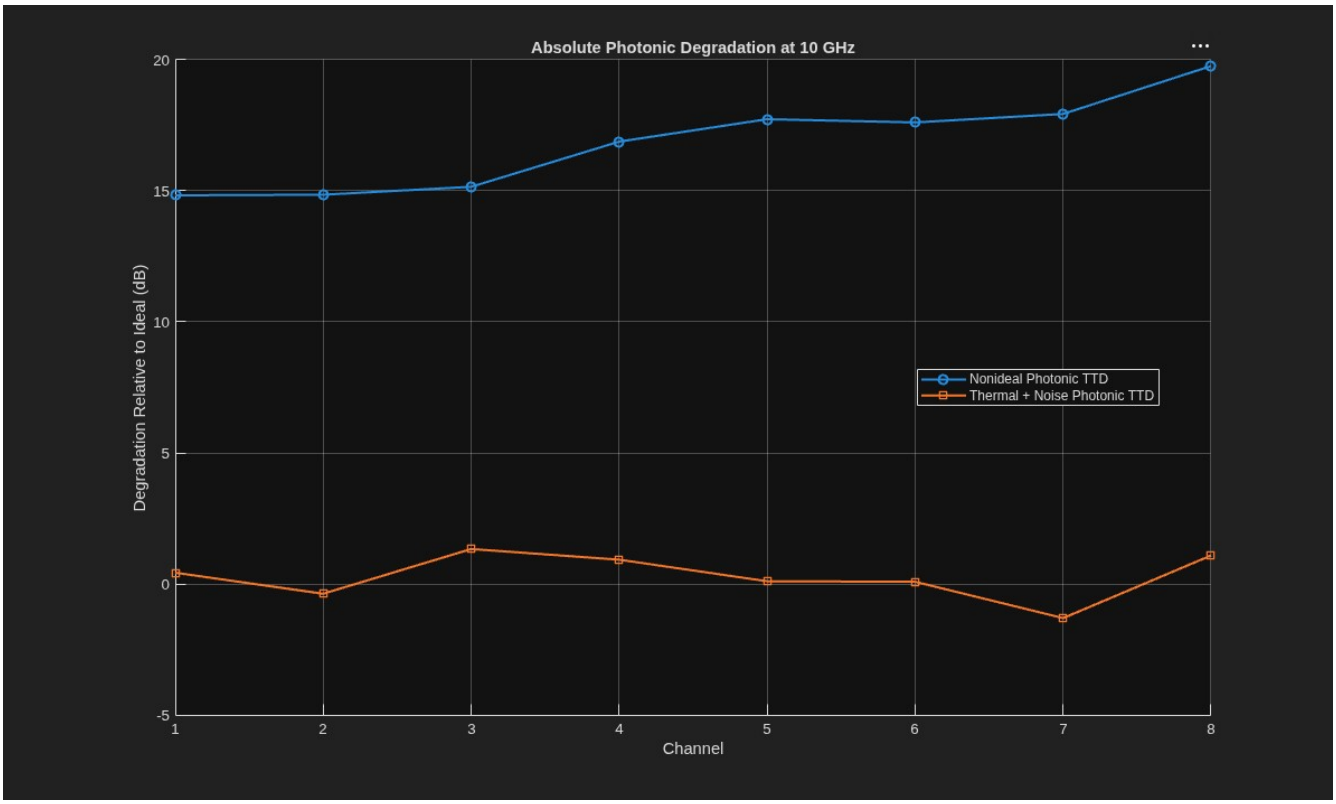


Figure 32: Absolute photonic degradation for non-ideal & thermal realizations

End-to-End Quantitative Comparison

The final quantitative results for a target steering angle of 30° across the 7.5–12.5 GHz frequency range are summarized below. Pointing error is evaluated relative to the target steering direction, while the delay-error and channel-imbalance metrics characterize the non-ideal photonic channels separately.

Table 2: End-to-end quantitative comparison for 30° steering

Metric	Phase Shifter	Ideal Photonic TTD	Non-Ideal Photonic TTD	Thermal + Noise TTD
Maximum pointing error	11.47°	0.000°	0.590°	1.310°
Channel imbalance at 10 GHz	N/A	0 dB	4.934 dB	2.636 dB
Delay-error MAE	N/A	approximately 0 ps	5.4509 ps	0.8535 ps
Delay-error RMS	N/A	approximately 0 ps	7.1766 ps	1.0776 ps
Worst-case delay error	N/A	approximately 0 ps	12.5772 ps	1.8733 ps

The fixed phase shifter presents the largest wideband degradation, reaching a maximum pointing error of 11.47°, while the ideal photonic TTD remains at the target angle to numerical precision. The non-

ideal photonic TTD limits the maximum pointing error to 0.59°, despite a maximum channel imbalance of 4.934 dB and a worst-case differential delay error of 12.5772 ps.

When thermal and noise effects are included, the maximum pointing error increases to 1.310°, with the largest deviation occurring at 8.600 GHz. The thermal and noise case shows a lower fitted delay error than the non-ideal case, but a larger pointing error. This is not contradictory, because the delay-error metric is extracted from the overall phase slope and therefore represents the effective differential delay, while thermal and noise effects can also introduce frequency-dependent phase and magnitude perturbations that degrade the instantaneous beam pattern without producing a large systematic delay-slope error.

The overall result is that both photonic TTD implementations maintain substantially better wideband pointing performance than the fixed phase shifter. However, the non-ideal, thermal and noise cases show that practical photonic TTD performance also depends on channel balance, differential delay accuracy and frequency-dependent phase stability.

PART 5: LIMITATIONS & PROJECT SUMMARY

Limitations

The RF part is an array-factor model. Except for the first total-factor example, it does not include full-wave element patterns, mutual coupling, feed-network effects or measured antenna S-parameters. The results therefore characterize the beamforming architecture rather than a complete physical antenna.

The INTERCONNECT network is also a system-level model. Noise is disabled in the ideal and non-ideal cases so that complex transfer functions and delay errors can be extracted cleanly. The non-ideal model is one randomized channel-length-error realization rather than a statistical fabrication model. The separate thermal + noise case combines a 20 K temperature increase with detector thermal noise, so it should be interpreted as a combined stress case rather than as an isolated thermo-optic validation.

Finally, normalized beam patterns remove common gain or loss. Insertion loss and relative channel imbalance must therefore be considered separately from pointing error and sidelobe behavior.

Project Summary

To sum up, this project covers a large part of the foundational concepts of RF and optical beamforming and connects them in a single system-level workflow. The MATLAB model first shows why fixed phase shifters work well around the design frequency but squint over wide bandwidth, while ideal TTD keeps the beam direction constant. INTERCONNECT then replaces the ideal delays with an 8-channel RF-to-optical-to-RF network, and the resulting H_n(f) responses are imported back into MATLAB.

For a 30° target angle over approximately 7.5–12.5 GHz, the fixed phase shifter reached 11.47° maximum pointing error, compared with 0.59° for the non-ideal photonic TTD and 1.31° for the thermal and noise case. The non-ideal implementation produced a delay-error MAE of 5.4509 ps, an RMS value of 7.1766 ps and a worst-case error of 12.5772 ps, while the thermal and noise case produced corresponding values of 0.8535 ps, 1.0776 ps and 1.8733 ps. At 10 GHz, the maximum

channel imbalance was 4.934 dB for the non-ideal case and 2.636 dB for the thermal and noise case. The main conclusion is therefore that TTD strongly reduces wideband pointing error compared with fixed phase-shifter beamforming, while practical photonic implementation remains sensitive to delay accuracy, channel balance and frequency-dependent phase perturbations.

Future work could expand the same model without changing its scope: use measured or MODE-derived frequency-dependent waveguide data, separate thermal and detector-noise effects into independent sweeps, introduce statistical and correlated fabrication tolerances across many photonic realizations, and connect the system-level array factor to a full-wave antenna model. These additions would strengthen the physical realism, but they are not required for the present phase shifter versus TTD comparison.

PART 6: REFERENCES

[1] C. A. Balanis, Antenna Theory: Analysis and Design, 4th ed., Wiley, 2016.

[2] P. Delos, B. Broughton and J. Kraft, "Phased Array Antenna Patterns-Part 1: Linear Array Beam Characteristics and Array Factor," Analog Dialogue, vol. 54, no. 2, May 2020.

[3] P. Delos, B. Broughton and J. Kraft, "Phased Array Antenna Patterns-Part 2: Grating Lobes and Beam Squint," Analog Dialogue, vol. 54, no. 2, June 2020.

[4] P. Delos, B. Broughton and J. Kraft, "Phased Array Antenna Patterns-Part 3: Sidelobes and Tapering," Analog Dialogue, vol. 54, no. 3, July 2020.

[5] J. Capmany and D. Novak, "Microwave photonics combines two worlds," Nature Photonics, vol. 1, pp. 319-330, 2007, doi: 10.1038/nphoton.2007.89.

[6] Ansys Optics, "Mach-Zehnder Modulator (MZM) - INTERCONNECT Element," Ansys Lumerical documentation.

[7] Ansys Optics, "Multimode Waveguide (WGD) - INTERCONNECT Element," Ansys Lumerical documentation.

[8] Ansys Optics, "PIN Photodetector (PIN) - INTERCONNECT Element," Ansys Lumerical documentation.